



DAVINCI 200

Maintenance Manual

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Introduction

The Davinci is an automated system for measuring critical dimension (CD), overlay or registration (OVL, RG) and/or film thickness (FT) on 150mm and/or 200mm wafers. It can be equipped with SMIF and/or open cassette (OC) loading options. The system is configurable with various options based on customer requirements or order specifications.

This document provides information on safety guidelines and procedures for performing preventive maintenance of the system.

Additionally operating instructions such as the User Manual, Preventive Maintenance Checklist and Safety Test also contain all the necessary information for proper and safe operation of the system. Please read and understand these documents thoroughly to ensure your safety and the safe use of the system.

About MueTec

MueTec develops high precision optical measurement and inspection solutions that help customers in the semiconductor and microelectronics industries to improve their manufacturing yield and overall process control. Our team of technical experts knows, fulfills and anticipates the needs of the global producers of wafers, masks and MEMS and offers custom-tailored and standard solutions that distinguish themselves through their extremely low maintenance requirements. Scalable systems allow for a combination of many different applications in a single system, including critical dimension, overlay and film thickness measurement as well as defect inspection and review.

In the last 25 years, MueTec has installed more than 300 measurement and inspection systems worldwide. There is hardly a leading semiconductor manufacturer that does not use our technology in production. Due to our ability to adapt systems precisely to customers' needs, our customers include not just large semiconductor companies, but also research institutions all around the world.

1 Safety Instructions

1.1 Personnel requirements

The operator, supervisor and service personnel for the DAVINCI system must be trained by MueTec training personnel or other knowledgeable and authorized persons in the operation and maintenance of the system before they do any work on or with the system themselves.

Furthermore, basic knowledge of the handling of wafers and safe behavior in a production environment and in a clean room are the prerequisite for the use of the system.

1.2 Labels on the System

The notes and warnings that are affixed on the system must be heeded carefully.

This label on the back of the system alerts users to the danger of live parts:



1.3 General Hazards and Precautions

To ensure safe maintenance, please read and observe the following instructions carefully.

1.3.1 General Safety Instructions



Only reach into the system when the maintenance manual provides for this and only in the manner described.



Do not touch moving parts.



Do not touch wafers under any circumstances.



Avoid shaking or moving the system.



The housing doors on the rear of the system may only be opened if the system is switched off (main switch off).



Connections for power supply and vacuum may not be made, manipulated or removed by the operator of the system.



Cables and lines on the system may not be connected, manipulated or removed by the operator.

1.3.2 Mechanical Hazard



Danger of injuries due to moving parts

The system has moving parts. Do not reach into the system or parts when parts are moving.

1.3.3 Electrical Hazard



Danger of electric shock

The system includes live components, cables and plugs. Keep hands and instruments away from wiring and connections which are electrically charged.

Electrical cabinets may only be opened by qualified, trained and experienced personnel, able to detect risks and to avoid hazards resulting from electricity. Also see norm EN 60204-1 – Safety of machinery – Electrical equipment of machines – Part 1: General requirements – Section 3.53.

For electrical work on the system, the specifications according to BGV A3 "Accident prevention regulation for electrical systems and tools" must be adhered to. For this system, the inspection schedule defined in §5 of BGV A3 have to be kept.

1.4 Special Safety Measures

The following notes and actions affect the entire life cycle of the system, from transport to commissioning and operation, on through to disposal.

The notes must be read and followed precisely by the operator of the system and by all people who come into contact with the system (supervisor, operator, maintenance and service personnel). Careful following the notes is absolutely necessary for proper and safe handling of the system.

1.4.1 Transport

- Before transport, the machine must absolutely be disconnected from power supplies. Take care that disconnecting does not create any electrical or fluid-connected hazards. Disconnecting must not be done during operation but only without voltage or pressure being applied.
- The set-up and dismantling, putting into service and disassembly must be done by specially-trained personnel. Staff must be informed about any potential risk involved in assembly and disassembly of the system. Staff has to have:
 - Knowledge of, a license for and qualification in transporting goods properly (e.g. crane operation, authorization for crane operation)
 - Knowledge of the system's load distribution, including that of machine components being transported.

If the required knowledge does not exist, it is preferable to contract a specialized company for this task. The company operating the system has the sole responsibility in this field of work.

- For the machine transport by separate load carrier (means of conveyance), it applies that: the transport of the machine and associated components may only be done with suitable vehicles, cranes, stackers or lift trucks (depending on the dimensions and weight of the system components). Devices must be sufficiently dimensioned to withstand the expected load and stress. Transport ways must be sufficiently dimensioned to avoid collisions with buildings, gateways or other machines.
- Lifting equipment used (cranes, ropes, attachment devices, etc.) must be sufficiently dimensioned for the total loads to be moved and be able to carry the expected loads. Take care of

lifting strictly horizontally. Be sure that the load is always stable. Devices may only be attached at the system's designated suspension points.

- Machine and machine components to be moved may only be lifted as high as is necessary for transport. Lifting excessively high increases the risk of dropping and damage.

1.4.2 Installation

- Installation and assembly of the machine and associated machine components may only be done by personnel named and appropriately trained by the manufacturer.
- The people will be informed about possible risks in direct consultation with the machine manufacturer and shall perform the duties assigned to them only after training. Be absolutely sure
 - that the personnel are comprehensively trained for installation and assembly,
 - that due rest periods are observed,
 - that only necessary, suitable and approved tools are used,
 - that personal protective equipment (PPE) is used by the operator whenever necessary or required by the actual circumstances on site.
- The personnel who will be performing installation and assembly work must be informed by the operator about risks that arise from the work environment or the planned location of the machine.

1.4.3 Commissioning

- The commissioning of the machine functions, associated settings (adjustments, parameters, programming) and necessary test runs in connection with the commissioning of the machine as well as individual machine components may only be performed by specialized personnel that has been named by the machine manufacturer and trained appropriately.
- The people will be informed about possible risks in direct consultation with the machine manufacturer and shall perform the duties assigned to them only after training. Be absolutely sure
 - that the personnel is comprehensively trained for installation and assembly,
 - that due rest periods are observed,
 - that only necessary, suitable and approved tools are used,
 - that personal protective equipment (PPE) is used by the operator whenever necessary or required by the actual circumstances on site.
- The personnel who will be performing installation and assembly work must be informed by the operator about risks that arise from the work environment or the planned location of the machine.

1.4.4 Occupational Health and Safety

- The operator must ensure that all people in constant contact with the machine receive sufficient rest periods and that a sufficient supply of fresh air is available to largely exclude signs of exhaustion and a lack of concentration when handling the machine.
- The operator must ensure that the work areas of the machine as well as the direct machine environment remain free of obstacles that could endanger the machine personnel. Tools needed for operation or maintenance have to be placed at their intended spots or at least in a position that causes no immediate danger. Unavoidable risks are to be clearly indicated by warning tape, signs, etc.
- Escape paths must be marked in every case and kept free of objects that could hinder escape.

1.4.5 Operation

- Before beginning any activity in the work area around the machine, the personnel must ensure that the floor conditions meet the requirements and that the work environment is safe to enter. Always make sure that

- spilled liquids, aerosols or other contaminations that can cause slipping are removed before work begins,
- the working space is sufficiently dimensioned to carry out work without collision or tripping,
- no unnecessary collision or tripping edges exist,
- unavoidable collision or tripping edges are covered and visibly marked.
- The area around the machine, especially the working areas, must be cleaned regularly. All necessary safety measures concerning cleaning must be provided by the operating company (e.g. signaling a slippery floor).
- The physical condition as well as the control-technical effect of the emergency stop controls on the entire machine must be checked at regular intervals in order to ensure the constant availability required by the Machinery Directive. Checks must be documented by the operating company. If the emergency stop devices are not in order, the system has to be shut down immediately. The machine may only be started up again after the emergency stop devices are fixed and working properly.

1.5 Emergencies

The system is equipped with an emergency off switch that disconnects the power to the system immediately and stops all movements and processes immediately.

In Case of an Emergency

Press down the emergency switch if there is ...

- There is a risk for people who are working on or near the system.
- a mechanical fault that could damage the system or the wafers.

Inform the responsible supervisor immediately. The checking and restart of the system may only be done by a supervisor.

In Case of a Malfunction



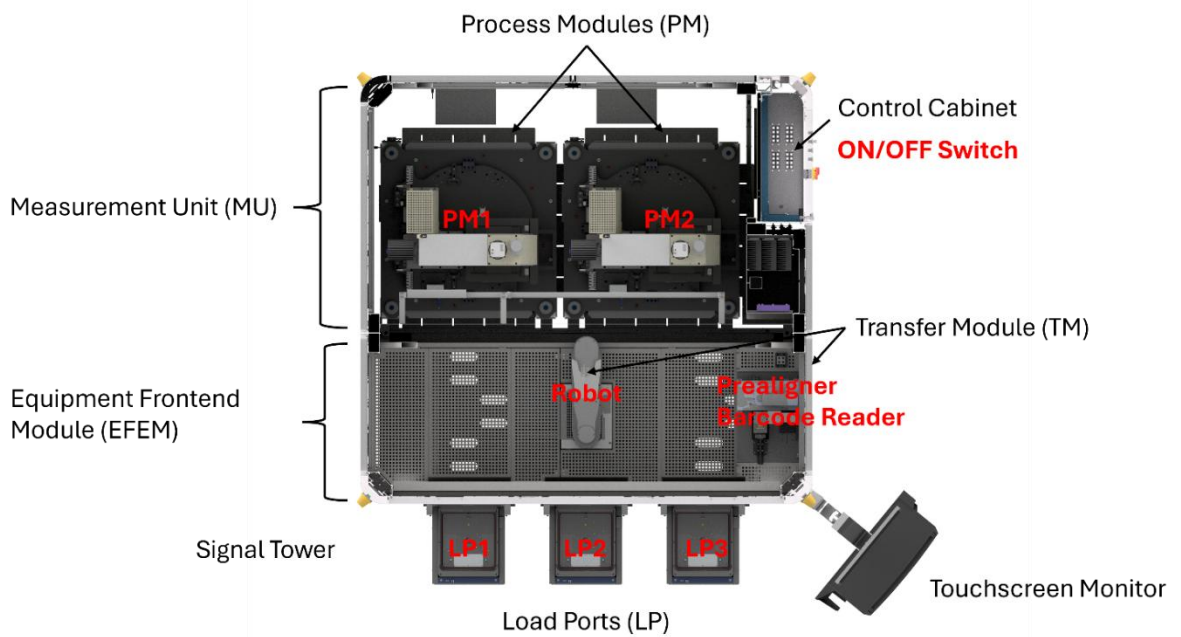
The housing doors on the rear of the system may only be opened if the system is switched off (main switch off).

2 System Description

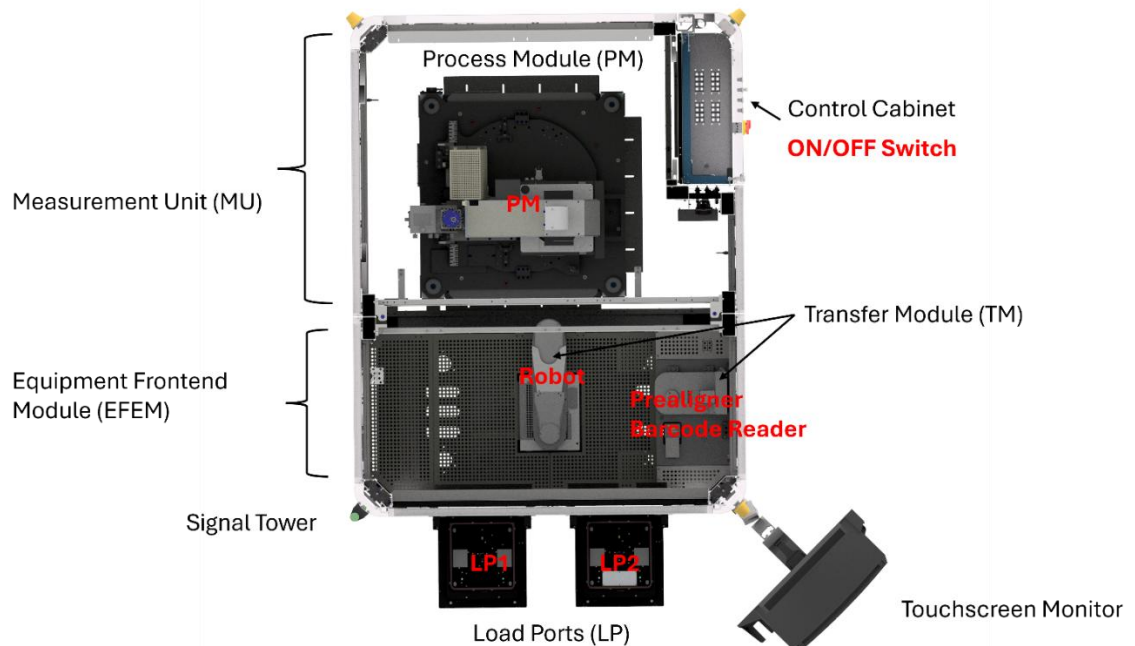
This section provides an overview of the essential components of the system, with a focus on the key components and operating elements. The number of process modules (PM) and loadports (LP) may vary depending on the system configuration.

2.1 Top View

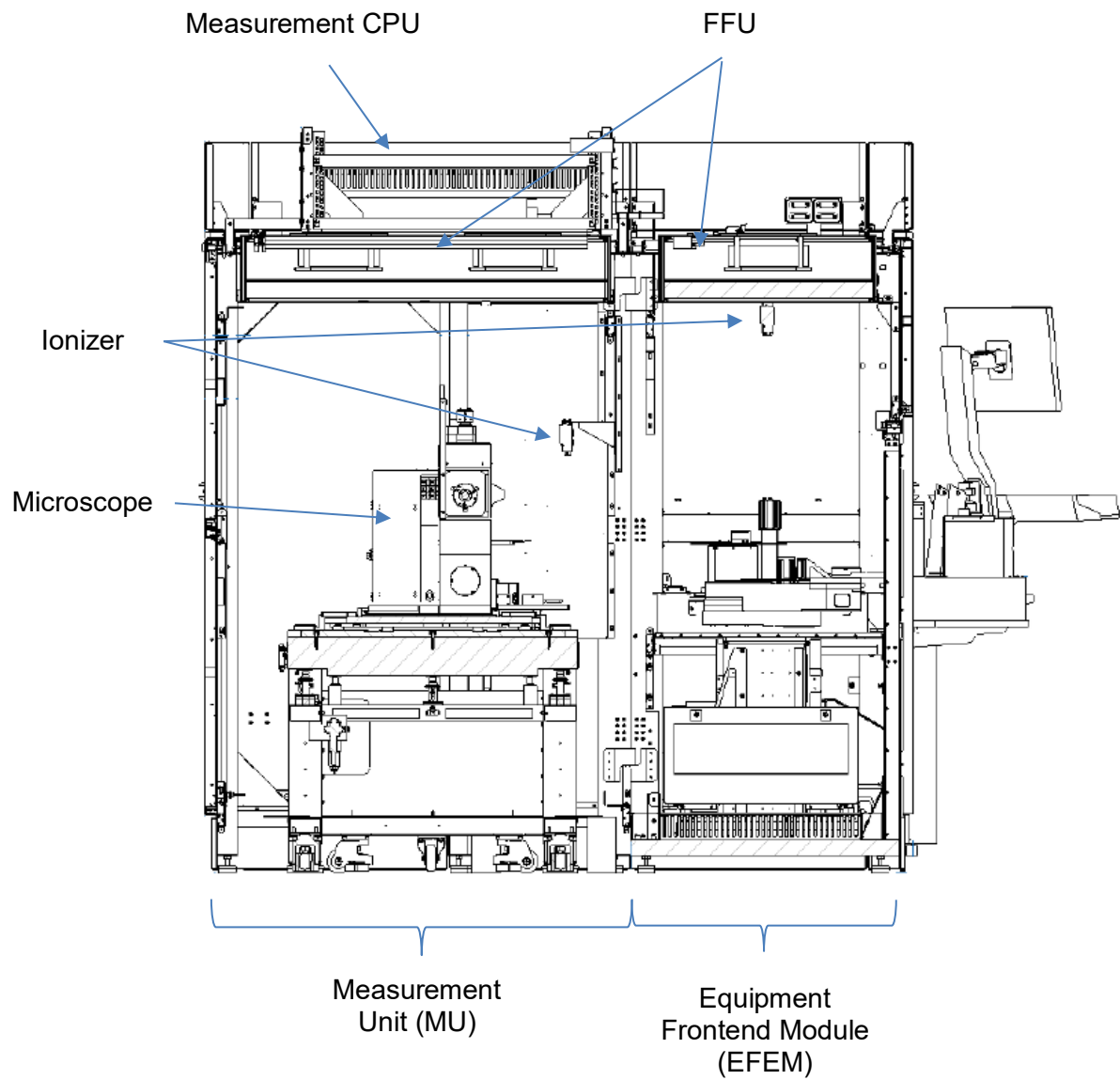
Example: 3 LPs, 2 PMs



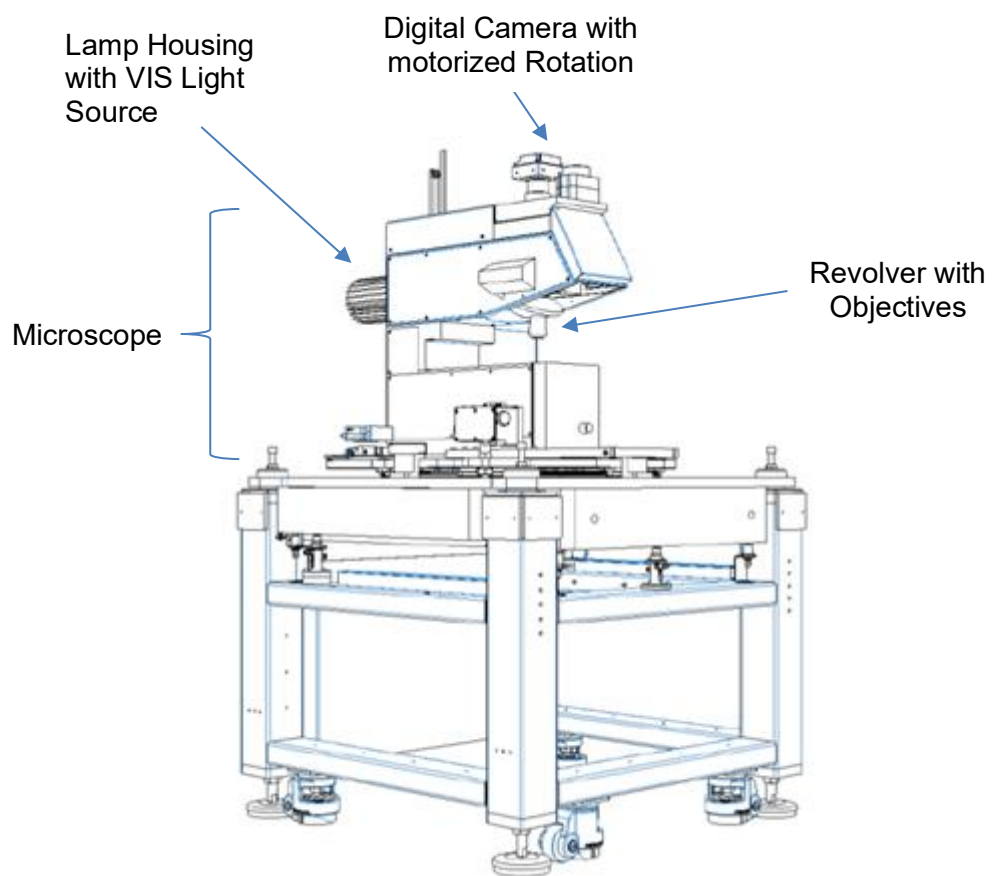
Example: 2 LPs, 1 PM



2.2 Side View



2.3 Process Module (Measurement Module)



3 PM Preparation

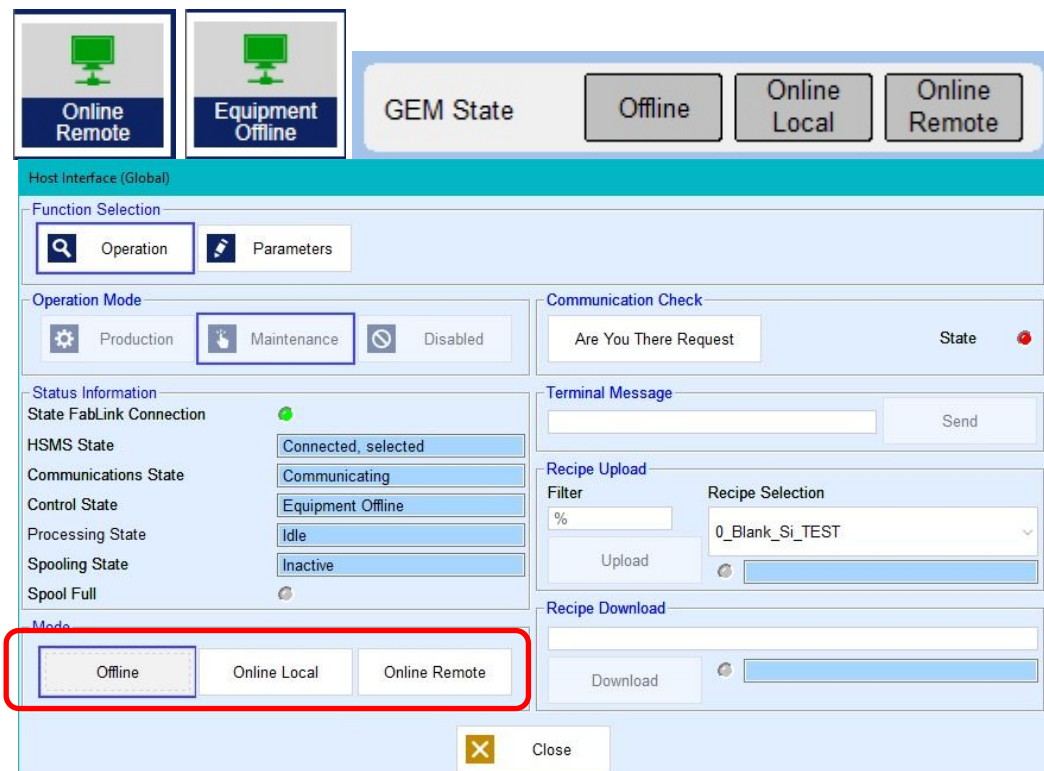
3.1 Preparation

3.1.1 Maintenance Duration

The annual maintenance takes approximately 1 day. To carry out maintenance successfully, certain components and materials must be prepared as in the DAVINCI - Preventive Maintenance Checklist.

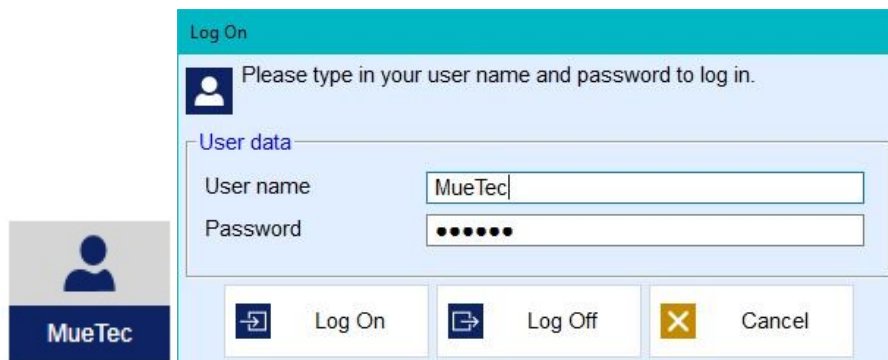
3.1.2 Host Interface

If the system is running in remote operation mode, change from remote operation to **Offline** or **Online Local** operation by pressing **Equipment Remote** button. Then select the **Offline** button and press **Close** button.



3.1.3 User Login

Have to login to administrator mode to run locally.

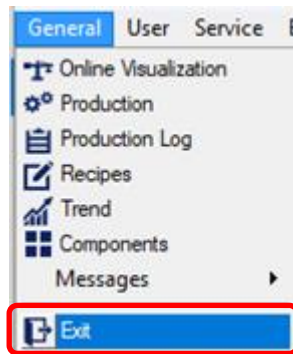


3.2 System Power Off and On

The system should only be switched off and on by the supervisor or an instructed person.

3.2.1 Power Off

1. Exit the **Tool Commander** either by
 - Selecting **General-Exit** on the Menu bar or



- Selecting the **Exit Program** button at bottom right



2. When the **Exit application** window is shown, tick the checkbox of **Shut down also the Control application** and press **Yes** button to exit the application.



3. After the **Tool Commander** & all other applications are closed, shut down the main PC.
4. After the main PC is shut down, press the **Main Switch (O)** button on the Control Cabinet door to switch off the system. The **Main Switch (O)** button illuminates when the system is powered off.



5. Turn the **Main Power Switch** which is located on the door of Control Cabinet in counterclockwise direction, to cut off the incoming power to the tool if necessary. (Please DON'T switch it on immediately after switching it off).



3.2.2 Power On

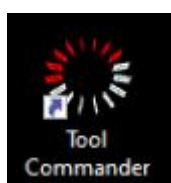
1. Turn the **Main Power Switch** which is located on the door of Control Cabinet in clockwise direction, to restore the incoming power to the tool if it is switched off. (30s waiting time after switching on is recommended before starting next steps).



2. Press the **Main Switch (I)** button next to the **Main Switch (O)** to switch on the system. The **Main Switch (I)** button illuminates when the system is powered on.



3. After the main PC is powered on and logged into Windows, start the **Tool Commander** (or named as **DaVinci**) or the service apps on desktop.



4 Robot

Robot or robot paddles may vary depending on the system configuration.

4.1 Robot Paddle

4.1.1 Check the Paddles

Visually check paddles for damage, especially near to the wafer contact area.

If contamination is found, clean the paddles with care. Do not rub hard the metal surface of paddles and do not touch the vacuum pads with the same wipe which was used for cleaning the metal surface. Gently wipe the metal surface of paddles with a new wipe.



4.1.2 Check the Vacuum Pads

Visually check the vacuum pads for damage, i.e. the three black pads on the paddles as shown in the right image above (depending on paddles design).

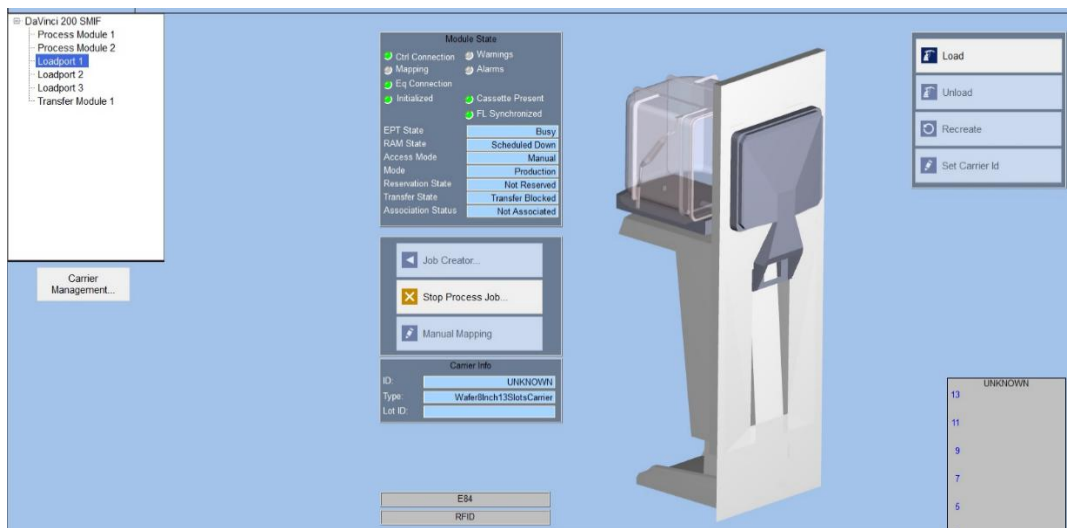
If contamination is found, clean the vacuum pads with care. Gently wipe the pads with a new wipe.

4.2 Robot Vacuum

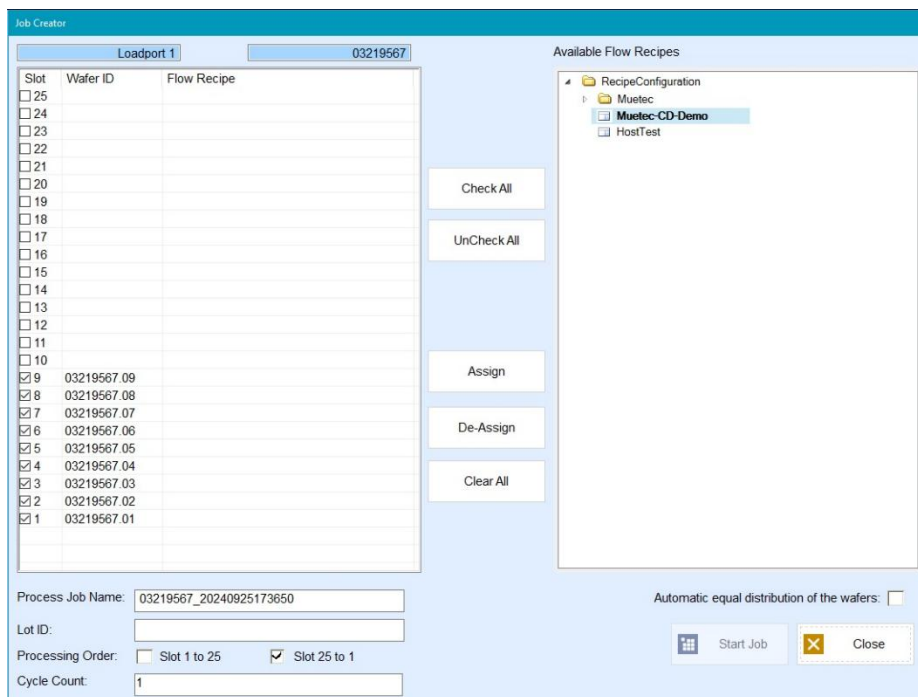
4.2.1 Check Vacuum Pressure

Check the vacuum pressure with and without holding wafer on paddles.

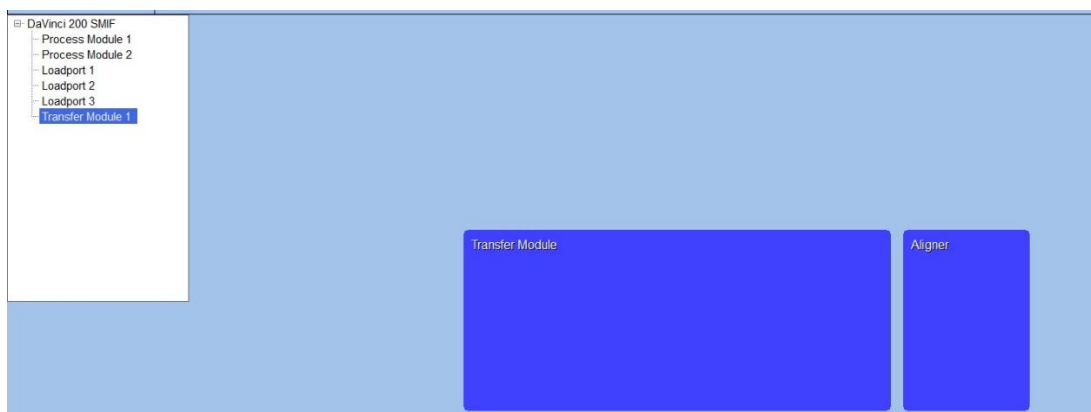
1. Load a SMIF/cassette/FOUP with the blank wafers onto Loadport1.
2. At AIS GUI, select **Loadport1** and press **Load** to load the wafers.



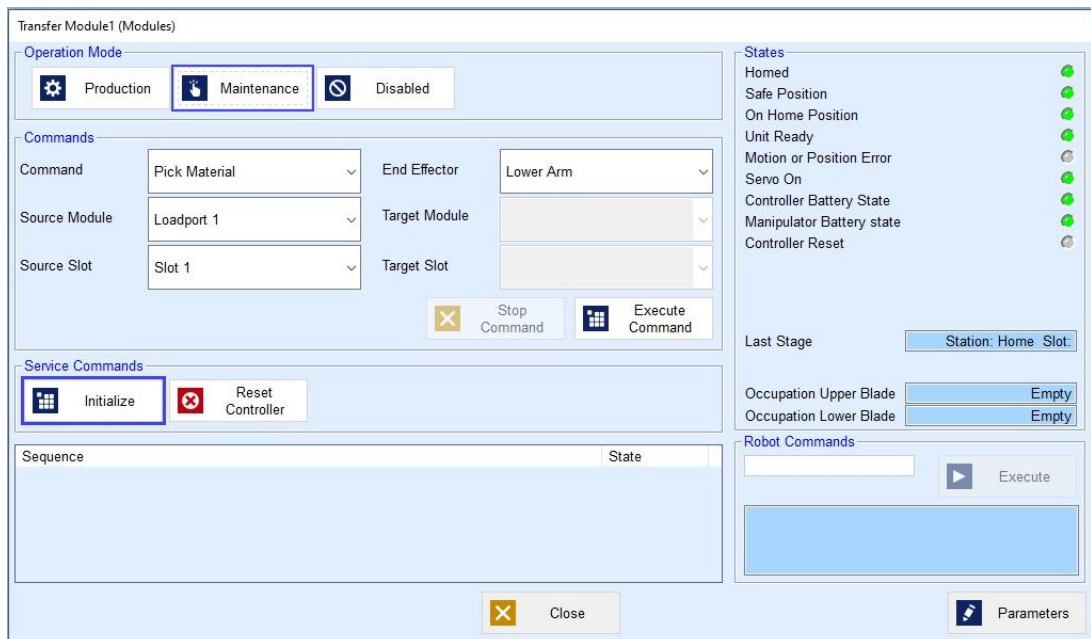
3. Close the **Job Creator** window after it popped out.



4. Select the **Transfer Module** in AIS GUI and select **Maintenance** as the operation mode.



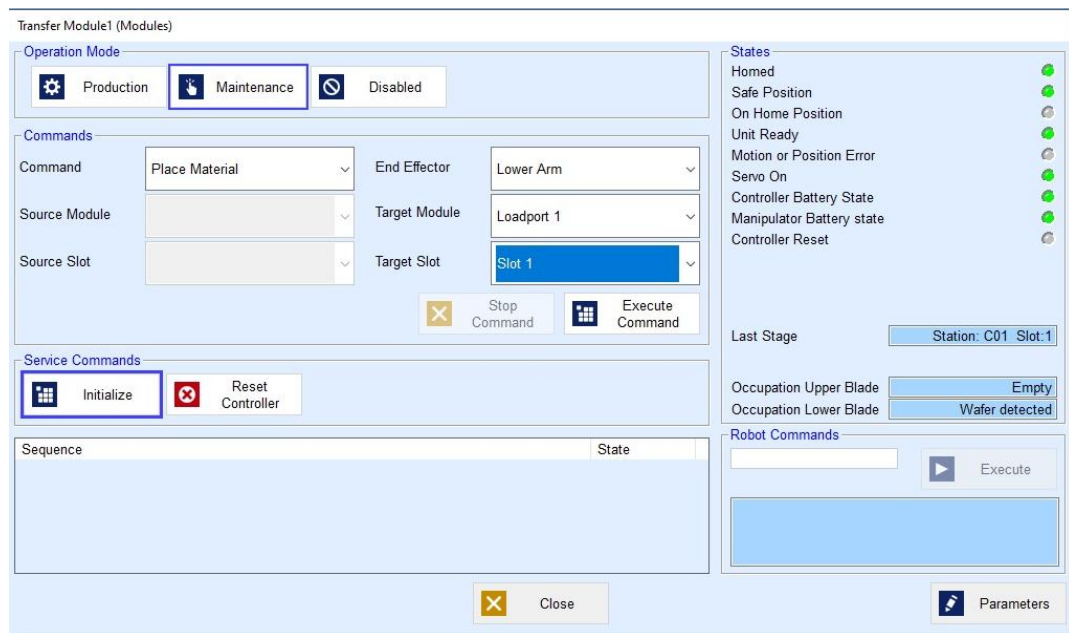
- Pick up a blank wafer by selecting **Pick Material** as command and **Lower Arm** as end effector to be used. Select the **Source Module** (Loadport01) and **Source Slot** (Slot01) where the wafer is to be picked up. Click **Execute command**.



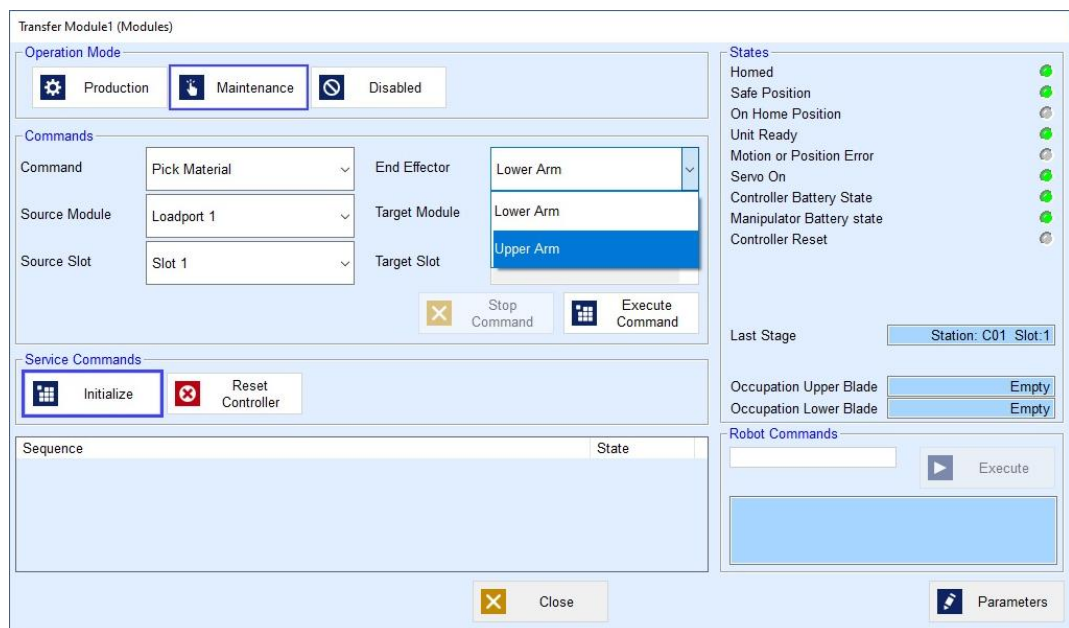
- Read and record the reading of vacuum pressure on the digital vacuum indicator of the robot. The unit is in kPa (depending on the type of robot).



- Place back the blank wafer by selecting **Place Material** as command and **Lower Arm** as end effector to be used. Select the **Target Module** (Loadport01) and **Target Slot** (Slot01) where the wafer was picked up from. Click **Execute command**.



8. After wafer placement, check if the reading is close to zero. Repeat for **Upper Arm** if installed.



4.2.2 Vacuum Valve

Check the vacuum valve by picking up and placement back of blank wafer (same as 4.2.1),

1. During picking up, monitor the vacuum pressure built up and the reading is shown on the digital vacuum indicator.
2. Check if the blank wafer is held securely on the robot paddles when vacuum is on.
3. During placement of the wafer, monitor the vacuum release. The reading should approach zero value.

4.2.3 Vacuum Lines

Visually inspect the vacuum lines for damage (ignore this if the vacuum lines are not reachable).

4.3 Robot Positions

4.3.1 Loadport 1, 2 and/or 3

1. Picking up a blank wafer (same as 4.2.1), check if the wafer edge is on the paddle mark.



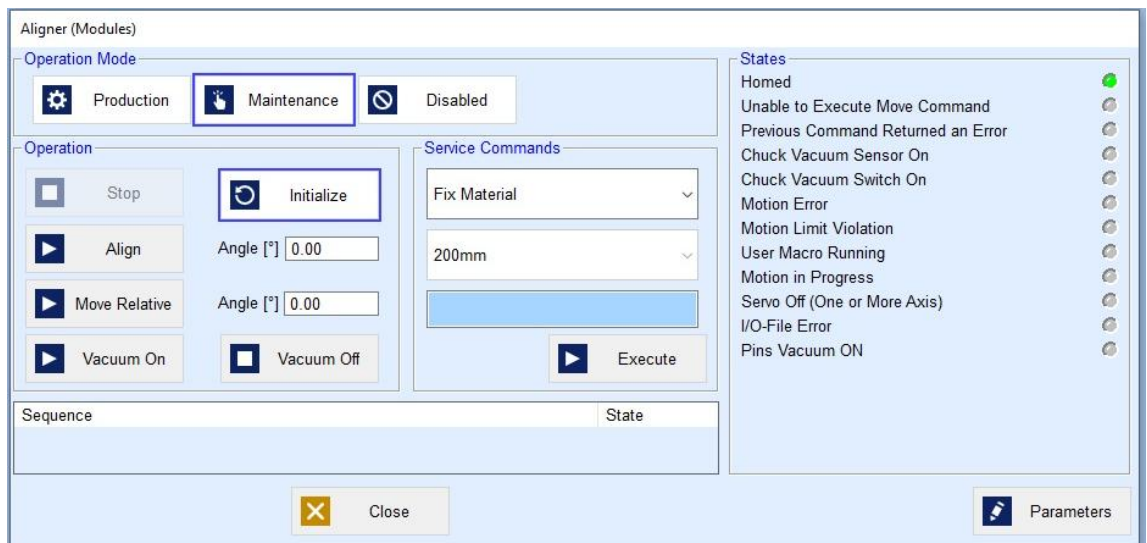
2. Place back the wafer to cassette and repeat for second paddle if installed.
3. Repeat the same checks for subsequent loadports.
4. Reteach the robot position if the wafer edge is way too far from the paddle mark.

4.3.2 Process Module 1 and/or 2

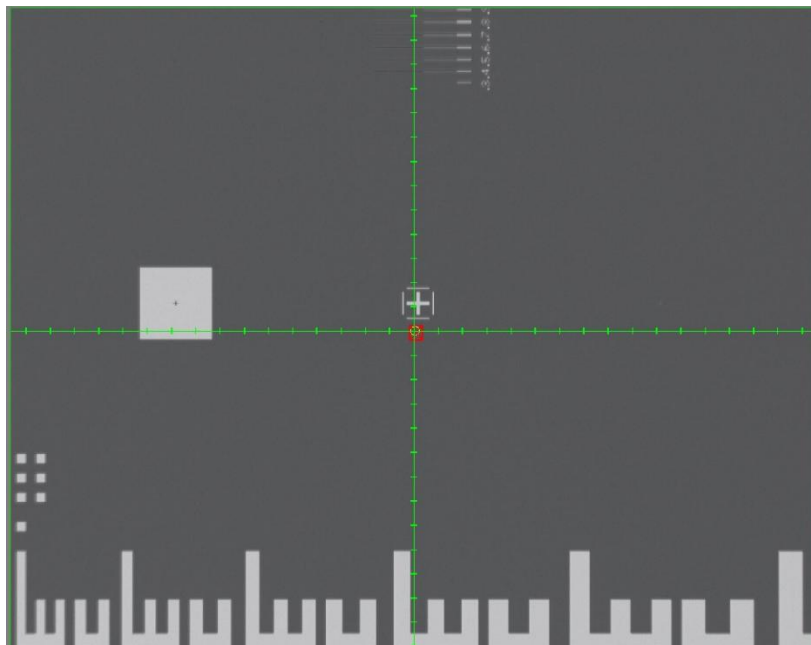
1. Picking up a system/blank wafer (same as 4.2.1),
2. Place the blank wafer to Aligner by selecting **Place Material** as command, **Lower Arm** as end effector to be used and **Aligner** as target module. Click **Execute command**.
3. After the wafer is placed on the Aligner, close the maintenance mode of **Transfer Module** and select the **Aligner Module**.



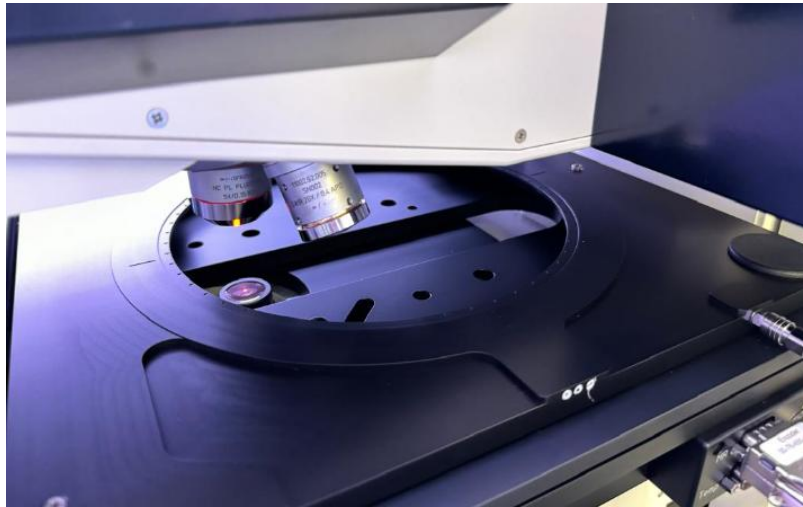
4. At the **Aligner Module**, select **Maintenance** as the operation mode.
5. At Service Commands, select **Fix Material** as command and click **Execute command** to vacuum the wafer.



6. At Alignment, click **Align** with zero degree to align the system wafer.
7. After alignment is done, at Service Commands, select **Release Material** as command and click **Execute command** to release the wafer.
8. Close the maintenance mode of **Aligner Module** and go back to the **Transfer Module**.
9. Before transferring to stage, be sure the objective used is 2.5x or 5x obj (low magnification).
10. Transfer the wafer from prealigner to stage by selecting **Transfer Material** as command, **Lower Arm** as end effector to be used and **Process Module1** as target module. Click **Execute command**.
11. After the wafer is placed on the chuck, drive the stage to wafer center at **STAGE-DRIVE-SUBSTRATECENTER**, check if the wafer center is close to the cross mark of system wafer.



12. If the installed chuck type holds the wafer by its edge, also check that the wafer fully covers all the vacuum holes on the chuck.



13. Reteach the robot position if the vacuum holes are not fully covered or vacuum leak.
14. Transfer the wafer from **Process Module1** back to **Aligner** by selecting **Transfer Material** as command, **Lower Arm** as end effector to be used and **Aligner** as target module. Click **Execute command**.
15. Realign the wafer and repeat for second paddle if installed.
16. Repeat the same checking for PM2 if installed.

4.4 Visual Inspection

4.4.1 Check Robot Movement

Visually check that the robot runs smoothly.

4.4.2 Check Robot

Visually check for damage to the robot

5 Loadport

The type and number of loadports may vary depending on the system configuration.

5.1 Visual Inspection

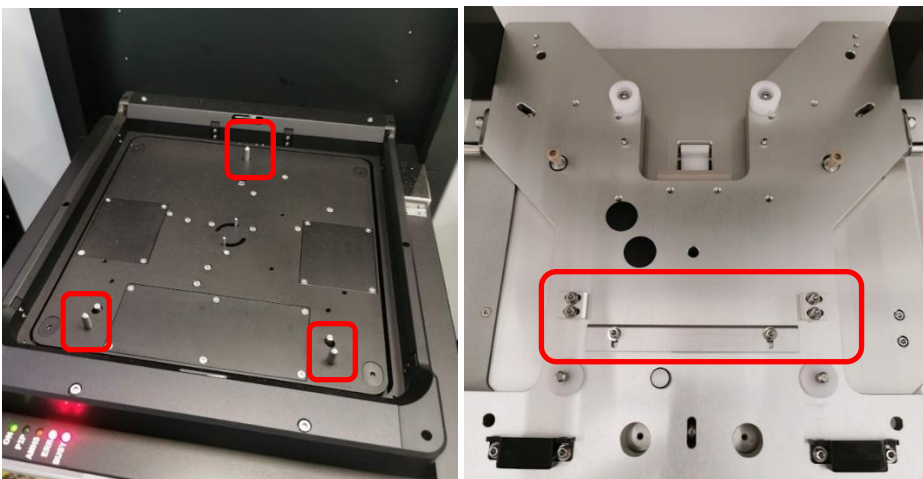
5.1.1 Check the Loadports

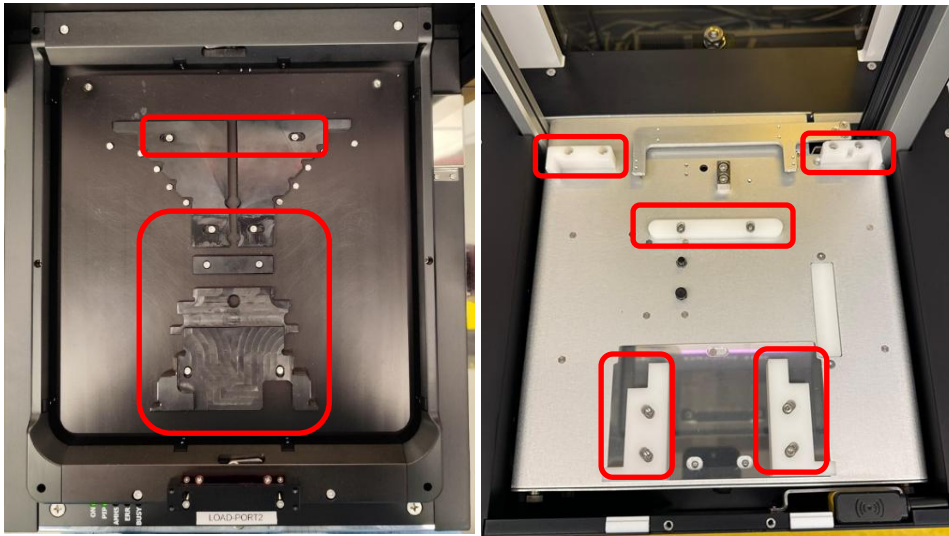
Visually check the Loadports for damage on the housing (may vary by design).



5.1.2 Check the Support Points

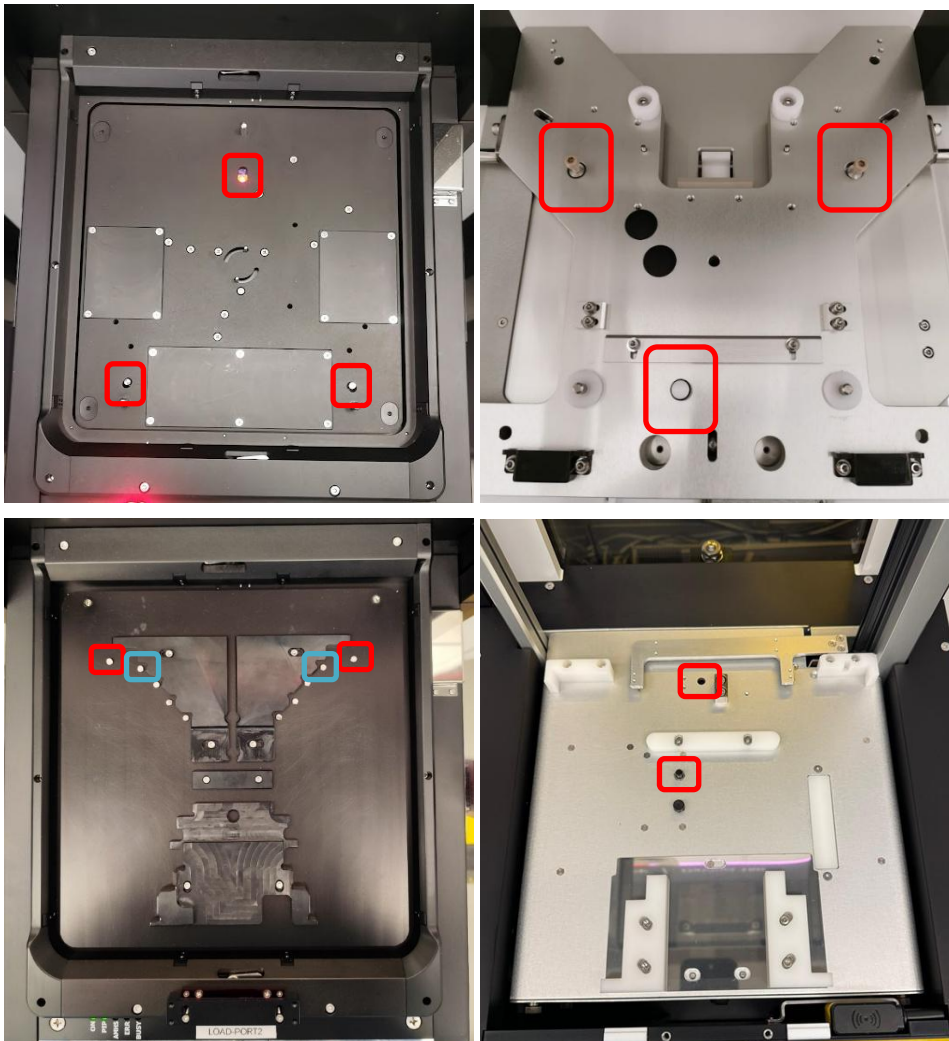
Check if the support points for the carrier/cassette is loosened or damaged (may vary by design).





5.1.3 Check the Sensors

Check if sensors for the carrier are damaged (may vary by design).

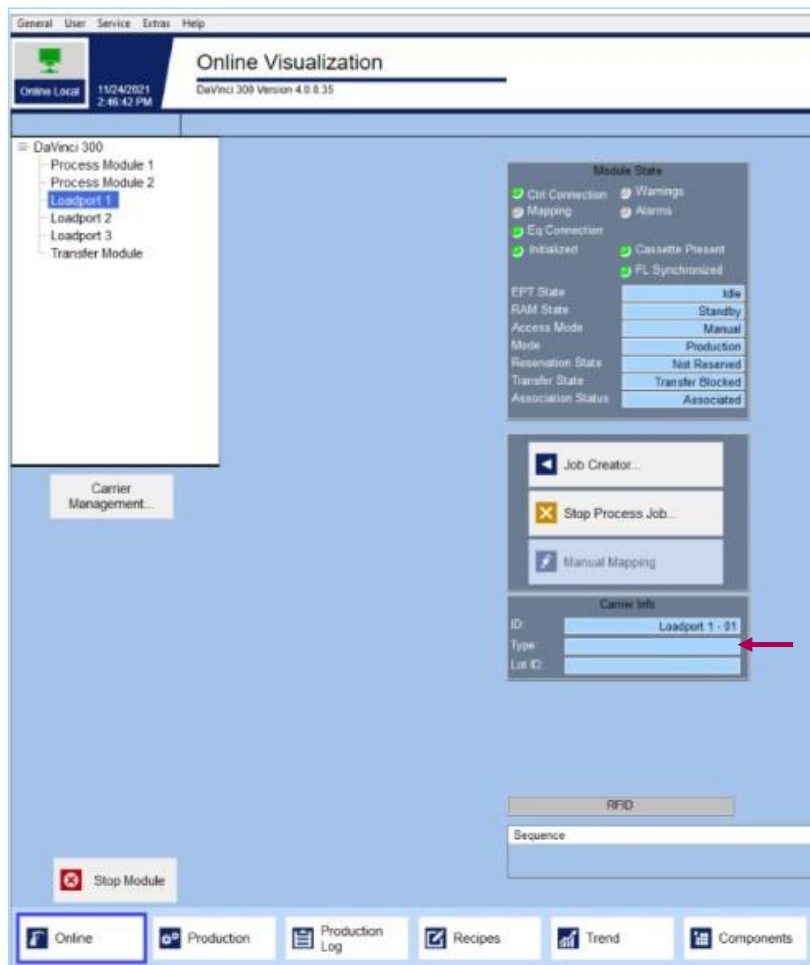


5.1.4 Carrier Size Detection Test (Open Cassette with Adapter)

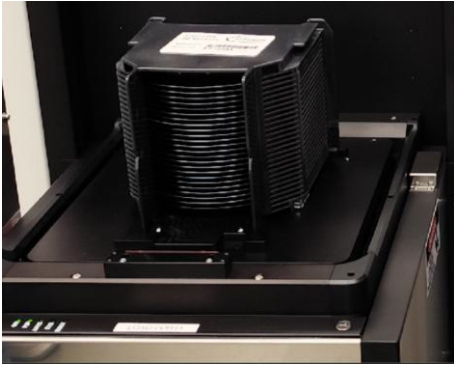
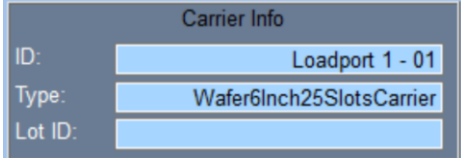
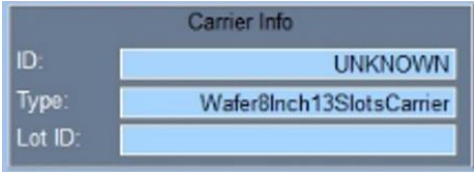
1. Start with no carrier on the loadports. Each loadports must be tested separately.



2. Load a cassette with blank wafers onto Loadport1 and at AIS GUI, select **Load Port 1**.
3. The carrier type is shown in the marked info window (also see table below):



4. Repeat the same checks for subsequent load ports.

Carrier Type	Info screen
None	
6inch 	
8 inch: 	

5.2 Manual Moving

5.2.1 Manual Loading of the Loadports

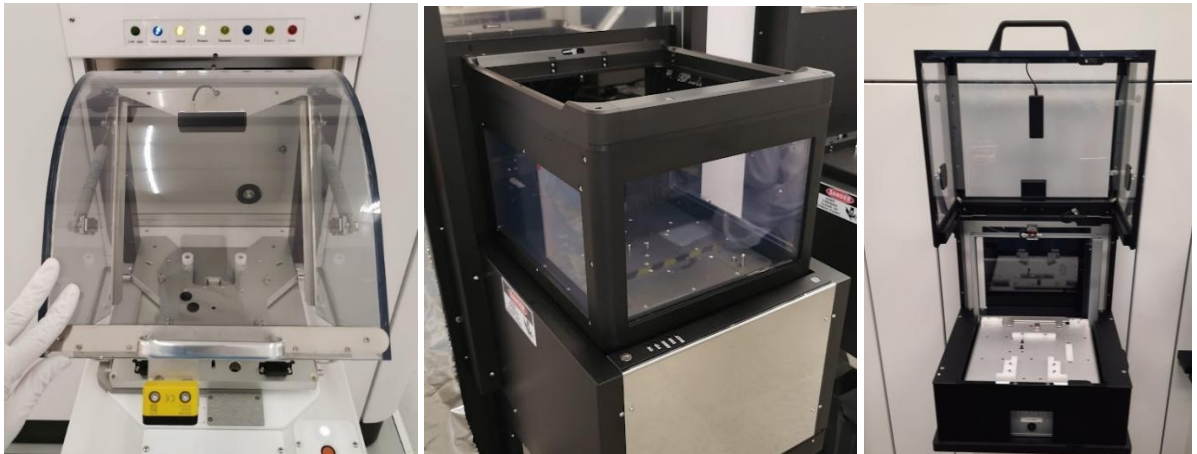
Load the Loadports manually with AIS TC and check for noise during movement.

5.2.2 Manual Unloading the Loadports

Unload the Loadports manually with AIS TC and check for noise during movement.

5.2.3 Check the Flaps

Check the flap (if installed) for opening and closing (may vary by design).



5.2.4 Check the Mapping Function

Check the mapping function with different kinds of wafer materials.

1. Load the SMIF/Cassette/FOUP with the wafers onto Loadport1.
2. At AIS GUI, select **Loadport1** and press **Load** to load the wafers.
3. Close the **Job Creator** window after it popped out.
4. Check if the mapping result is correct (mapping error message if double-slotted or cross-slotted wafers are detected)
5. At AIS GUI, press **Unload** to unload the wafers.
6. Repeat for other kinds of wafer materials.
7. Repeat for subsequent loadports.

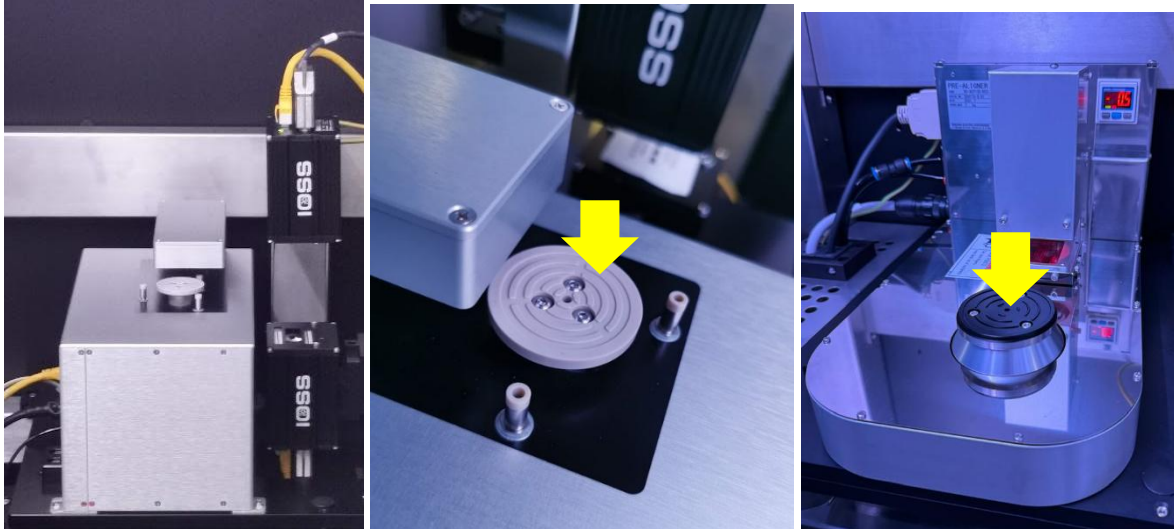
6 Prealigner

The prealigner may vary depending on the system configuration.

6.1 Visual Inspection

6.1.1 Check the Prealigner

Visually check the prealigner for damage (may vary by design)



6.1.2 Check the Contact Areas for the Wafer

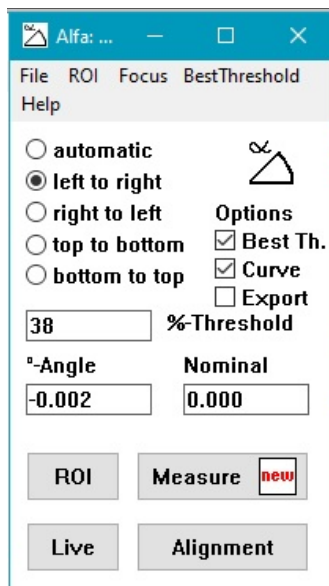
Visually check the contact areas for wafer (pointed by yellow arrow)

6.2 Manual Moving

6.2.1 Check the Alignment Accuracy

The pre-aligner default angle setting is used to setup the wafer notch facing to control cabinet direction when the wafer is placed on the stage. Check the alignment accuracy of the prealigner. Change the angle offset if necessary.

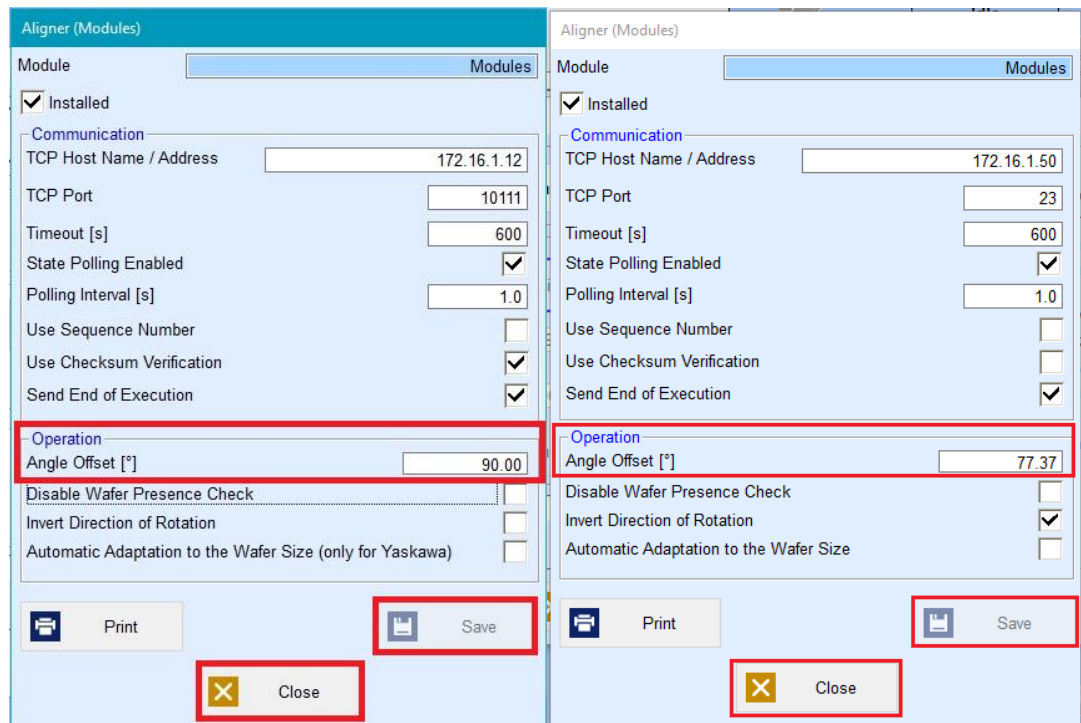
1. Load the SMIF/cassette/FOUP with the system wafers onto Loadport1.
2. At AIS GUI, transfer the system wafer to prealigner for alignment to zero degrees and then transfer the wafer to the stage of PM1. (same as 4.3.2),
3. At Nanostar, open **IP** and **Alfa** window. Measure a vertical or horizontal line with **Alfa** window.
4. Record the angle value shown on **Alfa** window. Check if the angle value $< 1^\circ$.



5. The new corrected angle is equal to the summation of the Alfa angle and with the current angle value. For example:

$$\begin{aligned}\text{New Angle (}^\circ\text{)} &= \text{Alfa Angle (}^\circ\text{)} + \text{Current Angle (}^\circ\text{)} \\ &= (-0.32^\circ) + 90^\circ \\ &= 89.68^\circ\end{aligned}$$

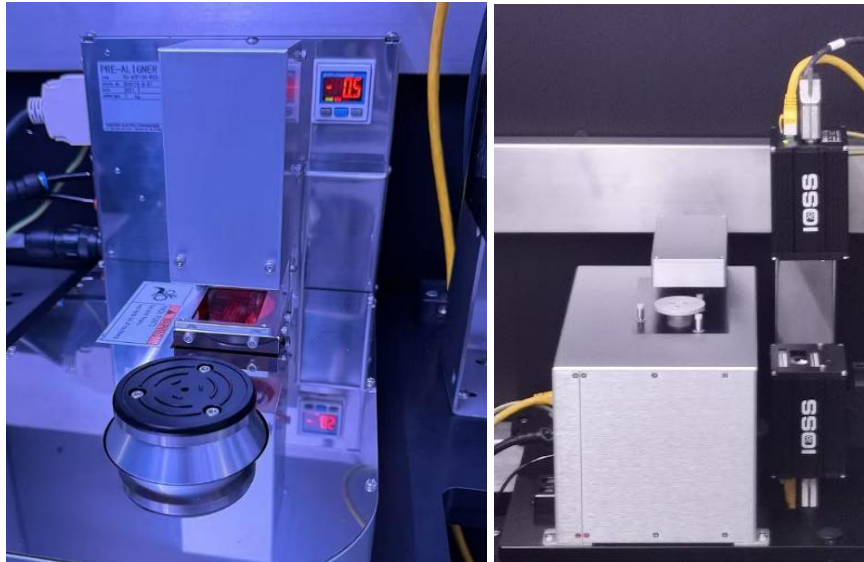
6. Fill in the new corrected angle into the parameter setting of the Aligner Module. Press **Save** and **Close** button. (setting may vary by system configuration)



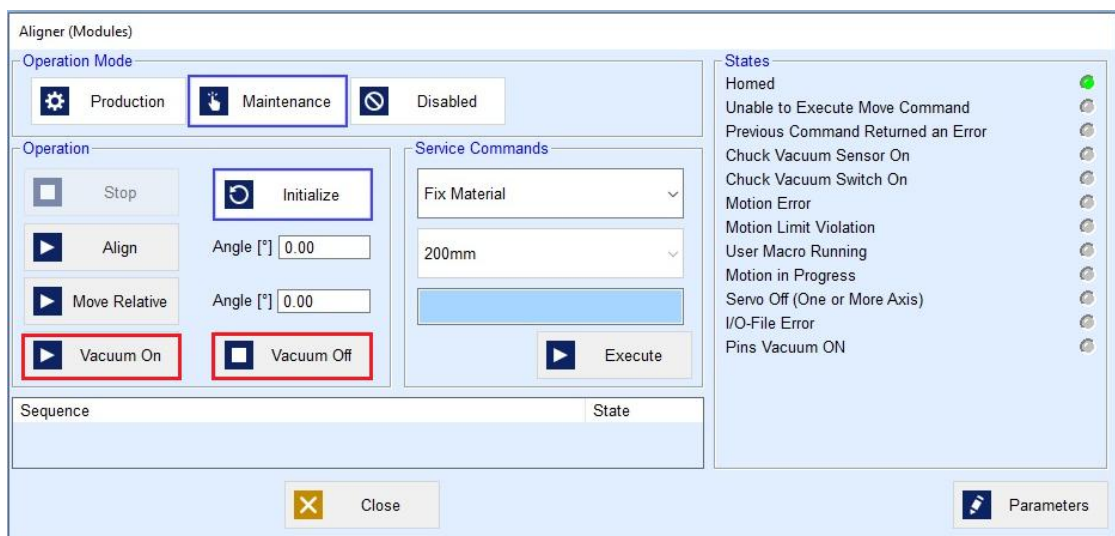
7. Transfer the wafer back to the prealigner and realign the wafer to zero degrees again before placing it back to stage. Check if the new correction angle value is $<1^\circ$.
8. Repeat for PM2 if installed.

6.2.2 Check the Vacuum on the Prealigner

Check the vacuum pressure by fetching the wafer with software, with and without holding wafer on prealigner



1. Pick up a blank wafer and place it at prealigner (same as 4.3.2),



2. At the **Aligner Module** window, click the **Vacuum ON** button to fix the wafer.
3. If the prealigner is with a vacuum meter, record the vacuum on reading.
4. If the prealigner is without vacuum meter, lightly push up the wafer to check if vacuum has built up.
5. At the **Aligner Module** window, click **Vacuum OFF** to release the wafer.
6. If the prealigner is with a vacuum meter, record the vacuum off reading.
7. If the prealigner is without a vacuum meter, lightly push up the wafer to check if vacuum is off.
8. Click the **Align** button to realign the wafer.
9. Place back the blank wafer by selecting **Place Material** as command and **Lower Arm** as end effector to be used. Select the target slot where the wafer was picked up from and click **Execute command**.

7 Barcode Reader

7.1 Frontside Reader (if installed)

7.1.1 Check Barcode Reader Position

1. At AIS GUI, click the **Components** button and select **Global** setting to look for **Reading Angle 150mm[°]** and/or **200mm[°]**. (depending on system configuration)

DaVinci 150OC (Global)

Component: DaVinci 150OC
Module: Global

Machine Name: Sample Application

Parameters:

Reading Angle 150mm [°]	0.00	Default Processing Order: Slot 25 to 1
Reading Angle 200mm [°]	349.00	
Reading Angle 300mm [°]	349.00	
Wafer ID Reader Reading Mode	Front side	

2. Pick up a blank wafer and place at prealigner (same as 4.3.2),
3. At **Service Commands**, select **Fix Material** as command and click **Execute command** to vacuum/fix the wafer.
4. Key in the reading angle in the **Align Angle entry field** and click **Align** to align the wafer for barcode reading position.

Aligner (Modules)

Operation Mode: Production, Maintenance, Disabled

Operation:

Stop, Align, Move Relative, Vacuum On, Vacuum Off

Service Commands:

Fix Material, 200mm, Execute

States:

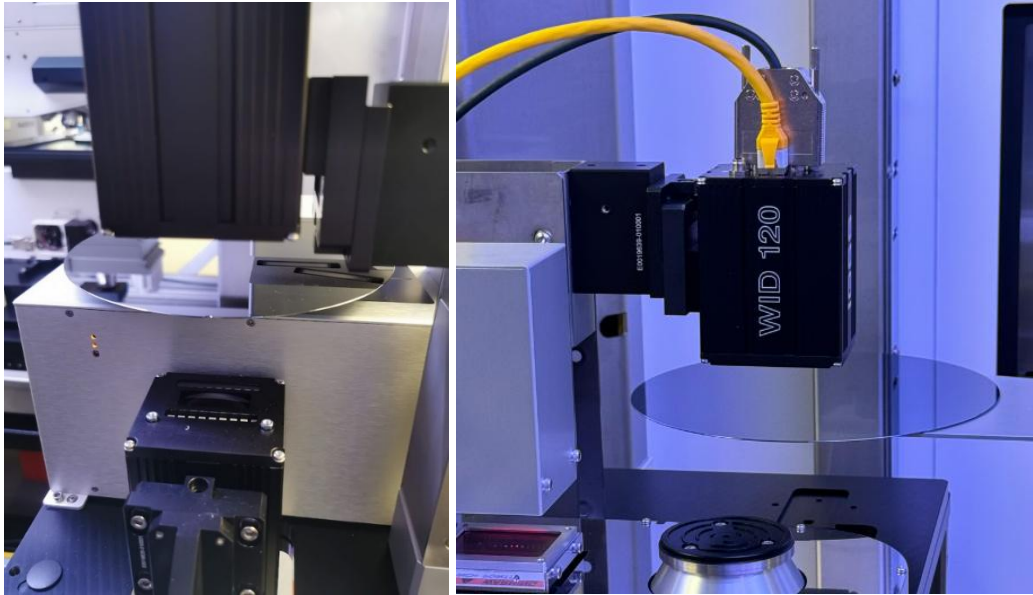
- Homed
- Unable to Execute Move Command
- Previous Command Returned an Error
- Chuck Vacuum Sensor On
- Chuck Vacuum Switch On
- Motion Error
- Motion Limit Violation
- User Macro Running
- Motion in Progress
- Servo Off (One or More Axis)
- I/O-File Error
- Pins Vacuum ON

Sequence, State

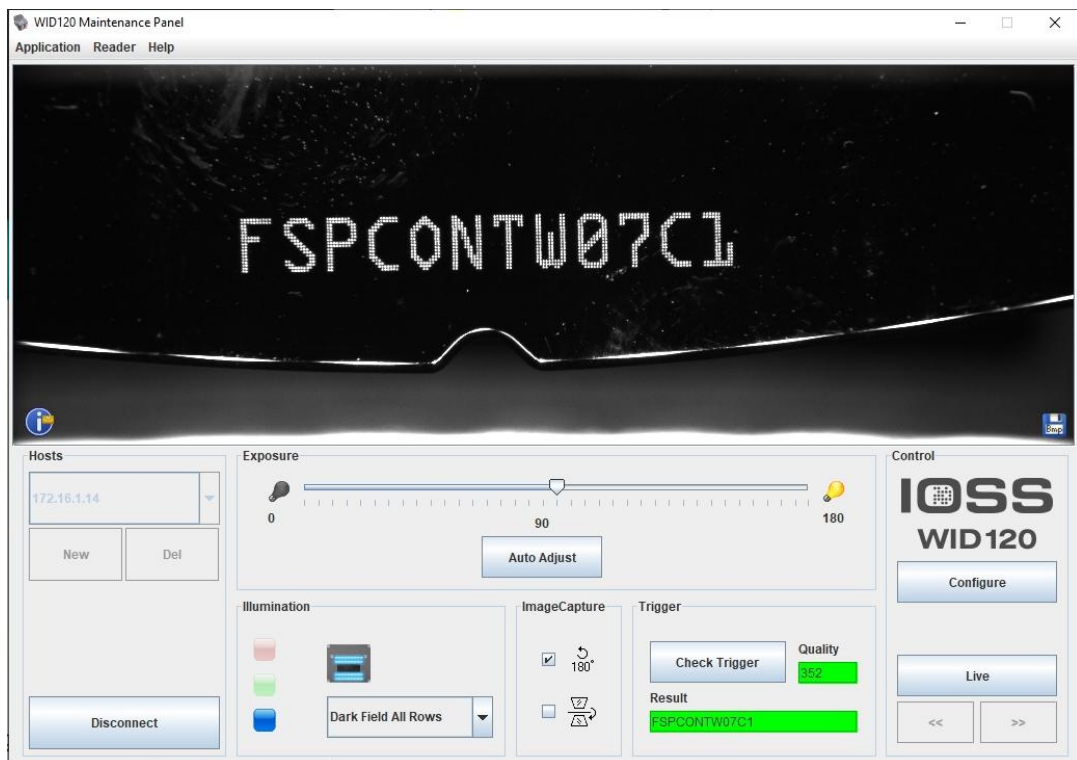
Close, Parameters

5. After alignment is done, at **Service Commands**, select **Release Material** as command and click **Execute command** to release the wafer.
6. Close the maintenance mode of **Aligner Module**, e.g. go back to production mode.
7. If the installed reader is another configuration where the reader cant read the waferID directly with the wafer on the prealigner, then will need to pick up the wafer with robot and place to the reader position.

8. Check if the Wafer ID is under the frontside reader position.

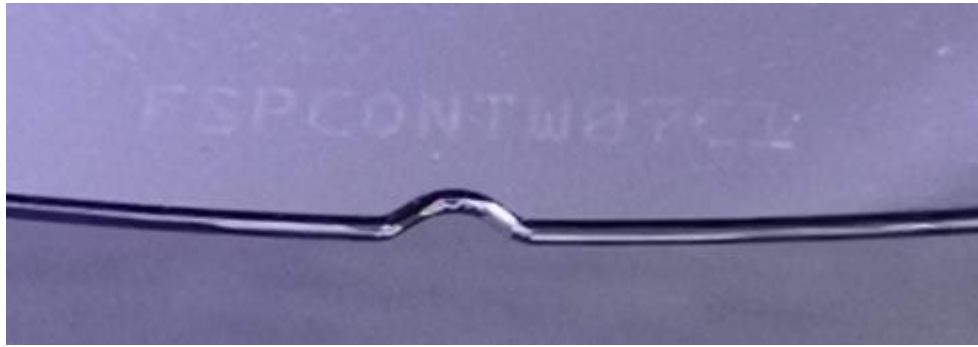


9. Launch the WID120 Maintenance Panel and connect to the Frontside Reader (172.16.1.14), check if the wafer ID is within the field of view.



7.1.2 Read Customer Wafer ID

1. At the WID120 Maintenance Panel, click **Check Trigger** to read the wafer ID.
2. Check if the readers able to read the customer wafer ID correctly.



3. Another method is to click the Barcode Reader Icon on the **Admin GUI**.
4. At the ID Reader, select **Maintenance** as the operation mode and click **Read Frontside**.

ID Reader (Global)

Operation Mode

☐ Production
 ☒ Maintenance
 ☐ Disabled

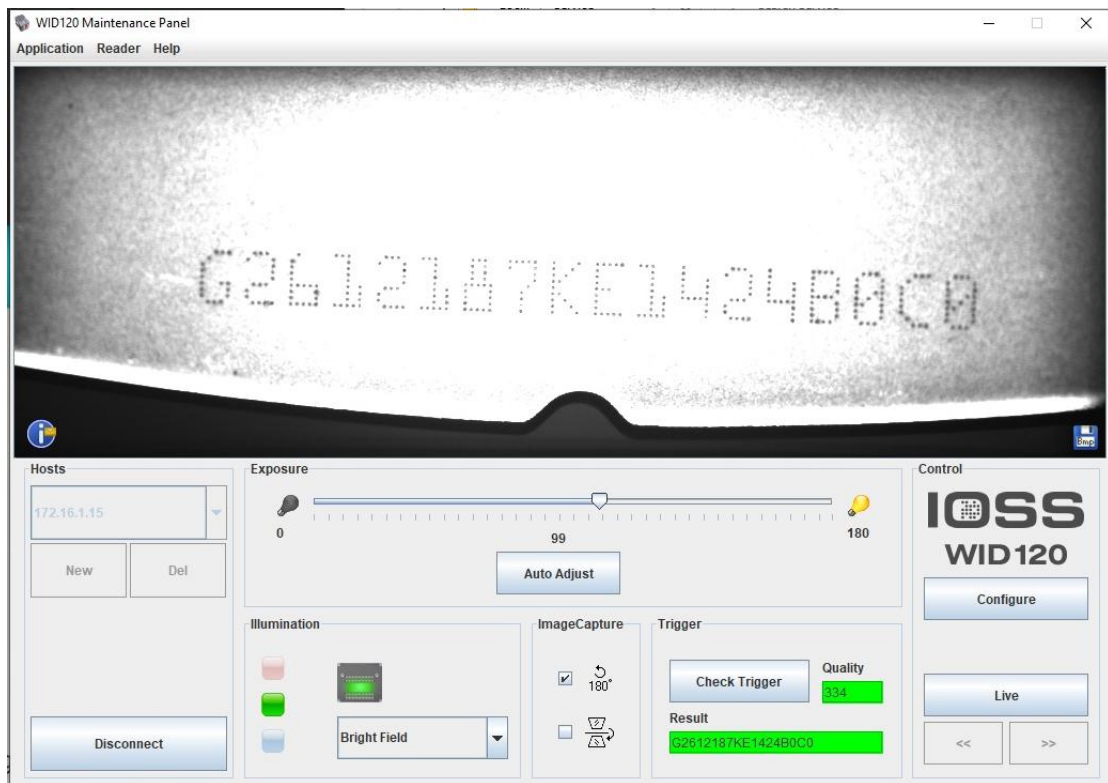
Read Frontside: FSPCONTW07C1
Result code: 2

Read Backside: G2612187KE1424B0C0
Result code: 2

7.2 Backside Reader (if installed)

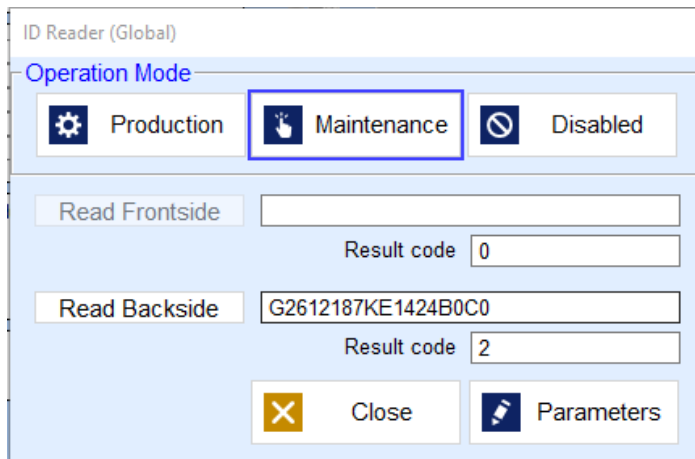
7.2.1 Check Barcode Reader Position

1. Check with the same method as the frontside reader
2. Launch the WID120 Maintenance Panel and connect to the Backside Reader (172.16.1.15), Check if the wafer ID is within the field of view.



7.2.2 Read Customer Wafer ID

Check with the same method as the frontside reader



8 Stage

Stage/chuck may vary depending on the system configuration.

8.1 Visual Inspection of the Stages and its components

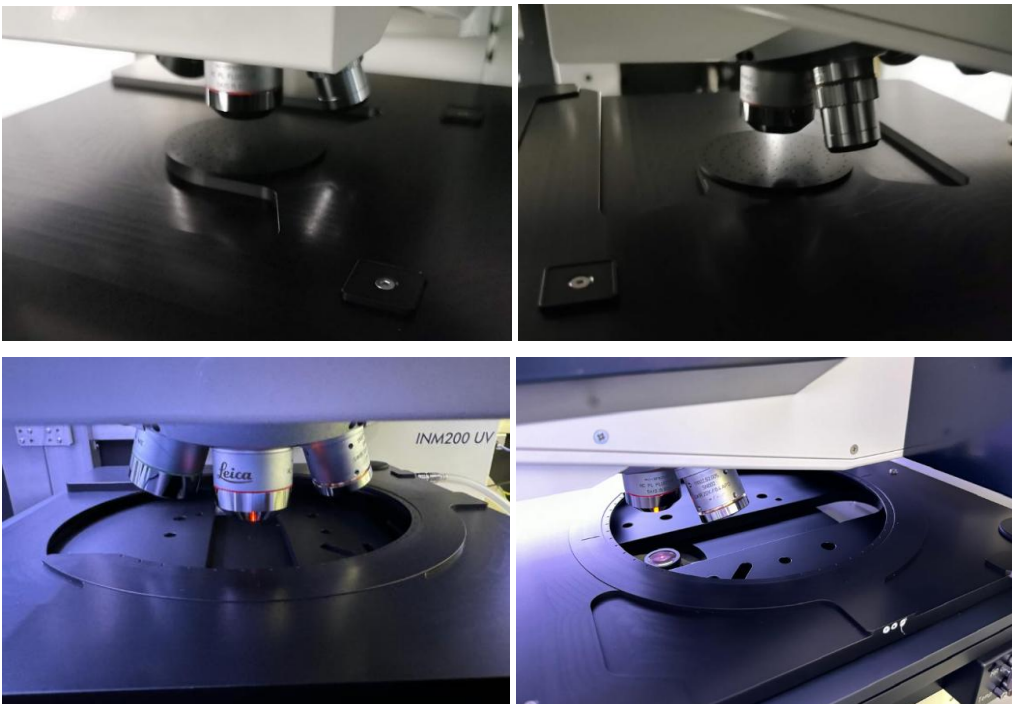
8.1.1 Check Stage

Visually check the stage for damage (depending on the system configuration).



8.1.2 Check Chuck Surface

Visually check the chuck surface for damage, scratches, resist and particles (depending on the system configuration).



8.1.3 Check the Cable Chain

Visually check the cable chain for damage if installed.

8.1.4 Check Stage Surface

Visually check the contact area of the stage for scratches.

8.2 Check Mechanic and Function of Process Modules

8.2.1 Check the Smooth Running of the Carriage

1. Click the **TASK-keypad** and click **Freemode**.
1. Drive in both X & Y direction.
2. Check if the carriage is running smoothly.
3. Repeat for PM2 if installed.

8.2.2 Check the Signal from the Position Measurement System

Check the signal from the position measurement system if installed.

8.2.3 Check the Robot Handover Position

Check the robot handover point (chuck load position) (Same as 4.3.2)

8.2.4 Check the Vacuum Signals on the Vacuum Controller

The vacuum controller is at the back of the measurement platform. Open the machine housing and place the bypass unit on the sensor.

Record the vacuum reading with and without a wafer placed on stage. Load a wafer on the stage by running a recipe or by manual loading in maintenance mode.

Typically, the VacuumWaferDetectedThreshold setting is -700mbar, if the vacuum reading is >0.700bar with a wafer on the stage, then it is acceptable. The VacuumBadThreshold setting is -400, if the vacuum reading is <0.400bar, it is detected as low vacuum. The vacuum thresholds may vary depending on the setup condition of customer facilities.



8.2.5 Check the Vacuum Signals on AIS GUI

On the main interface of AIS GUI, click **Admin GUI**. Record the vacuum reading with and without wafer on stage.

Main pressure [bar]:	0.0	
MP vacuum [mbar]:	0.0	
EFEM vacuum [mbar]:	0.0	
Vacuum Chuck A 12" [mbar]:	0.0	8": 0.0
Vacuum Chuck B 12" [mbar]:	0.0	8": 0.0

8.2.6 Check the Vacuum Guides

Visually check the vacuum guides if installed.

8.3 Cleaning the Process Modules

8.3.1 Clean the Stage

Only use the indicated detergents for cleaning. Other solvents may cause corrosion and oxidation of parts.

- Marble air float surface (if installed): check before operation (no oil, no particles). Use cleanroom wipes dipped in a small amount of anhydrous ethanol for one-way wipe.
- Grating tape (if installed): check the grating tape every 6 months for oil particles. Use cleanroom wipes dipped in a small amount of anhydrous ethanol to wipe.
- Sport cable (if installed): use cleanroom wipes dipped in a small amount of anhydrous ethanol to wipe the sport cables or use nitrogen for cleaning.
- Metal surface (if installed): use cleanroom wipes dipped in a small amount of anhydrous ethanol for cleaning.

8.3.2 Clean the Energy Chain

Clean the energy chains for PM1 and PM2 (if installed).

8.3.3 Clean the Guides

Clean the guides with dry cleanroom wipe for PM1 and PM2 (if installed). Gently wipe in one direction and use new wipes regularly. Do not use isopropanol or any other solvent to clean the linear guides. Otherwise rust might emerge.

8.3.4 Greasing the Linear Guides

Grease the linear guides with ISOFLEX NBU 15 grease for PM1 and PM2 (if installed). Do not use too much grease. Apply grease very decently and thinly to the V-shaped guides.



9 Optic

9.1 General Check of the Optic Components

9.1.1 Check the System for Damage

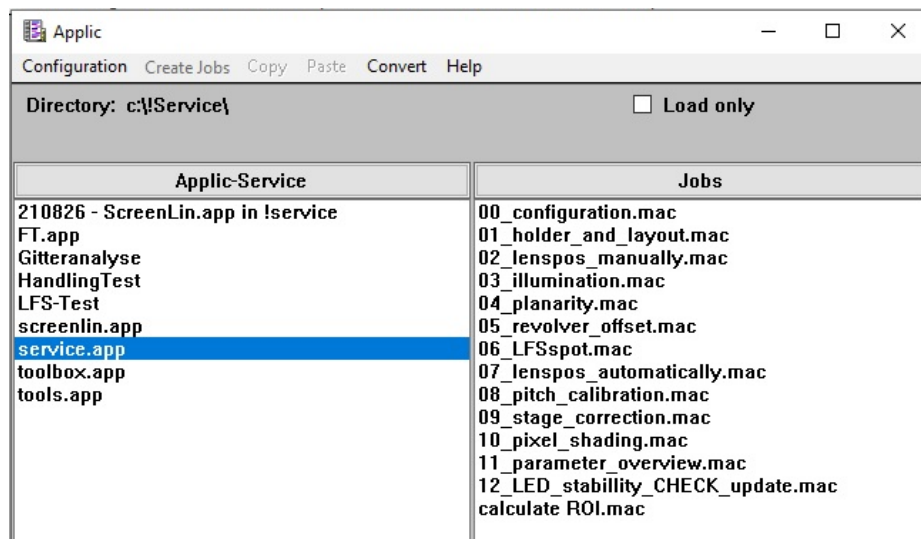
Visual check the optic system (microscope) for damage on PM1 and PM2 (if installed).

9.1.2 Check any Loose Components

Check that no components are loose.

No strong forces are recommended to delicate parts such as objectives, camera, etc.

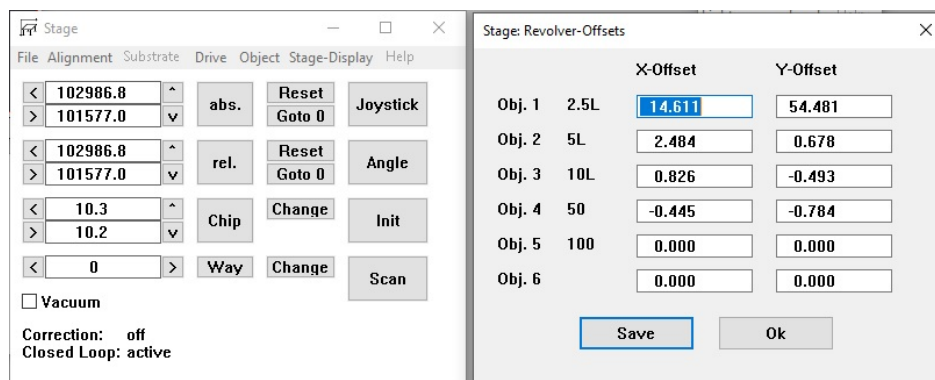
9.2 Service Macros



9.2.1 Objective Revolver Offset

Execute the Service Macro: **Objective Revolver Offset**. Check if the set values are correct.

1. Load the system wafer to stage of PM1 with the wafer notch/flat at zero degrees.
2. Take a screenshot of the stage revolver offset values at **TASK-STAGE-FILE-REVOLVER OFFSET** and save as a backup



3. In the menu of the **Nanostar** window, select **Start – Start Service**.
4. Select **Service.app** and run the **05_revolver_offset.mac** service macro.



5. In Docu Window of the **Revolver Offset**, click **F2: Alignment** to perform stage alignment.
 - Select obj 10x obj and reflective light mode.
 - During stage alignment, align the bottom left corner of the white box at upper right of the wafer.
6. After stage alignment is done, click **F5: Check Revolver Offset**.
7. The system will auto check the revolver offset, starting from 100x→50x→20x→10x→5x→2.5x objective (depending on the objectives installed)
8. After completed, click **F9: Show Protocol**. If the result is < 5um (pass), take a screenshot to save the result.

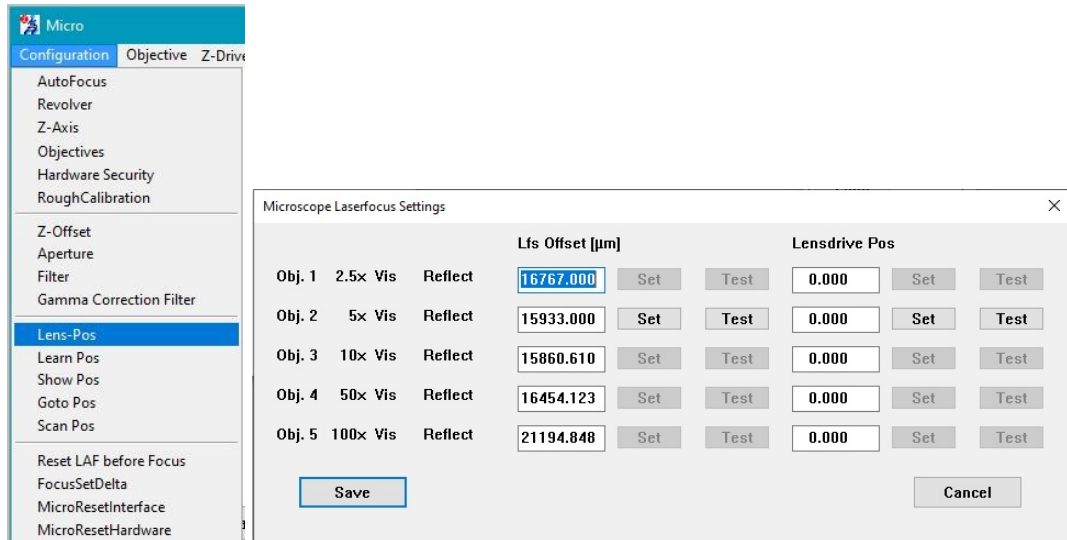
X- and Y-Offsets			
System	DaVinci		01.12.2022 16:09:51
Customer			
Serial Number	DaVinci		
Operator	Service007		
Obj.	Magnif.	Delta X (µm)	Delta Y (µm)
1	2.5	-0.2	-0.3
2	5.0	-0.4	0.5
3	10.0	-0.2	0.5
4	50.0	-0.6	0.2
5	100.0	0.0	0.0
6	0.0	0.0	0.0

9. If the result is >5um (fail), then a new offset must be learned.
 - a. Run the service macro-05_revolver_offset.mac.
 - b. Click **F2: Alignment** to perform stage alignment.
 - c. Click **F3: Set Offset to Zero** and check if the offset for all objectives is zero.
 - d. Click **F4: Learn New Revolver Offset**.
 - e. The system will auto learn & save the new offset.
 - f. Check the new offset at **TASK-STAGE-FILE-REVOLVER OFFSET**.
 - g. Rerun **F5: Check Revolver Offset** to check on the result again (<5um).
10. Repeat for PM2 if installed.

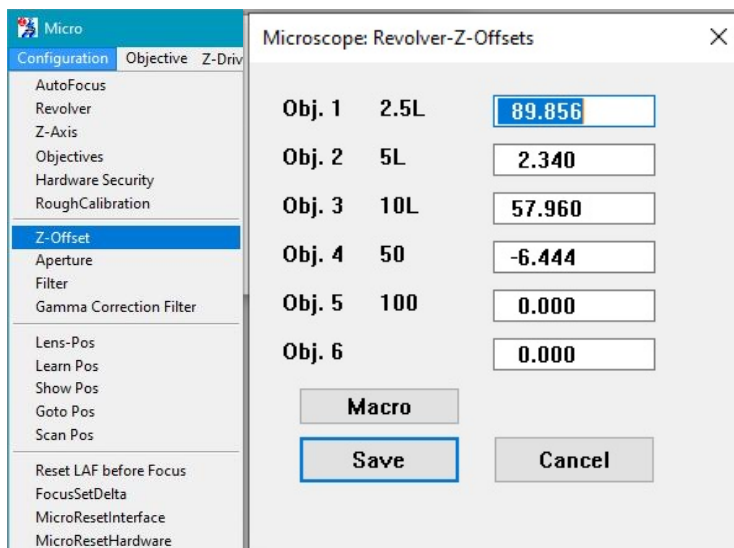
9.2.2 LFS-Offset

Check the Service Macro: **LFS-Offset**. Check if the set values are correct.

1. Load the system wafer to stage of PM1 with wafer notch/flat at zero degrees.
2. Take a screenshot of the laser focus setting at **TASK-MICRO-CONFIGURATION-LENSPOS-LFS OFFSET** and save as a backup.



3. Take a screenshot of the Revolver-Z-offset at **TASK-MICRO-CONFIGURATION-ZOFFSET** and save as a backup.

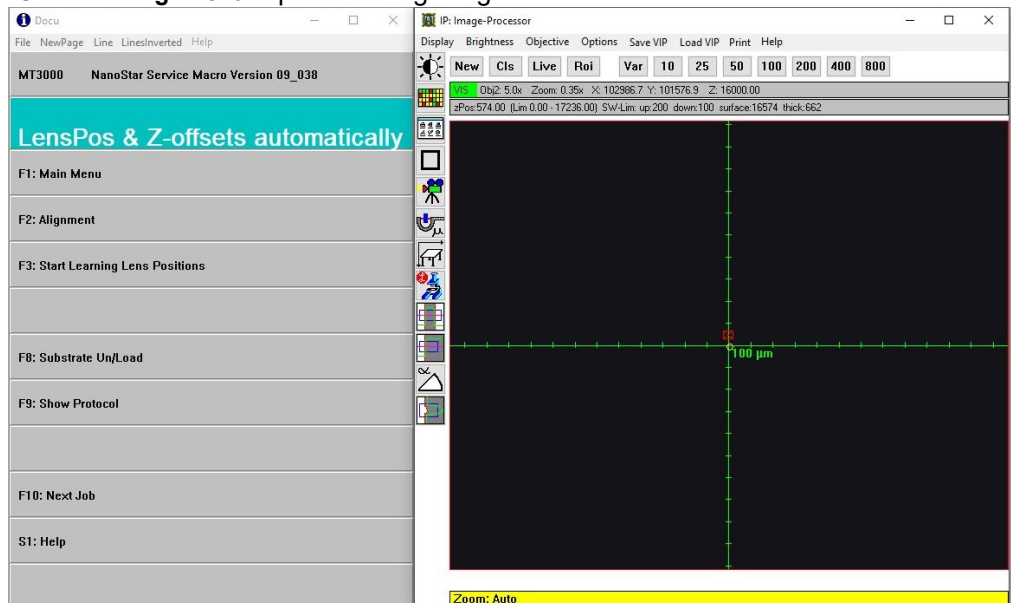


4. Perform the checks manually for all objectives one after another.
5. Open the **TASK-KEYPAD-FREEMODE**



6. Starting with the 2.5x objective, move to a white area with CD linewidth number using joystick (if installed) or mouse.
7. Click **Focus Off** on the keypad.
8. Adjust the Z to slightly defocus.
9. Click **Focus On** button on the keypad.
10. Check if the image becomes focused & sharp, take a screenshot & save the image.
11. Repeat the same check for all other objectives.
12. If the image is not sharp after **Focus On**, relearn the new offset.

- a. Run the service macro **07_lenspos_automatically**.
- b. Click **F2: Alignment** to perform stage alignment.



- c. Select 10x objective & reflective light mode.
 - d. During alignment, align the bottom left corner of the white box at the upper right of the wafer.
 - e. Click **F3: Start Learning Lens Positions**
 - f. Tick all 5 objectives or any specific objective that needs to update the LFS offset.
 - g. The system will auto learn and focus for the selected objectives.
 - h. After completed, save the new offsets.
 - i. Click **F9: Show Protocol**.
 - j. Recheck again after saving the new offset values.
13. Repeat for PM2 if installed.

10 Control Cabinet

10.1 Visual Inspection

10.1.1 Check any Loose Components

Check that no components are loose, e.g. loose wires and apparent damage.



10.1.2 Check PLC

Check the PLC for any apparent errors.



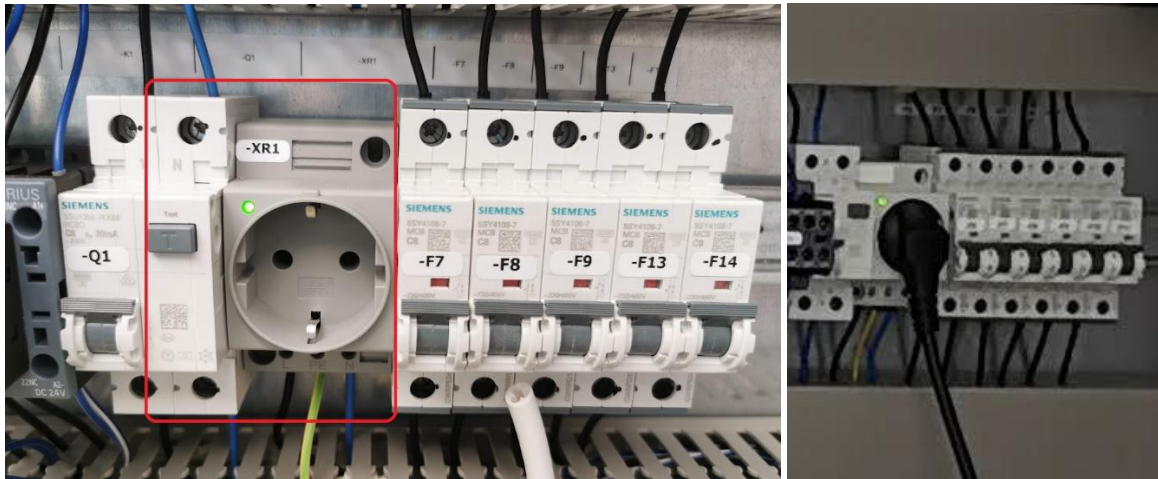
10.2 Check fuse

10.2.1 Test the FI

Test the FI (if installed), circuit breaker for the service socket.

Press the test button and the LED on XR1 will be off.

Press again and the LED will turn back to green.



10.2.2 Test the Fuses

Check if the fuses loosened from the clamp or if there is apparent damage.



11 Housing

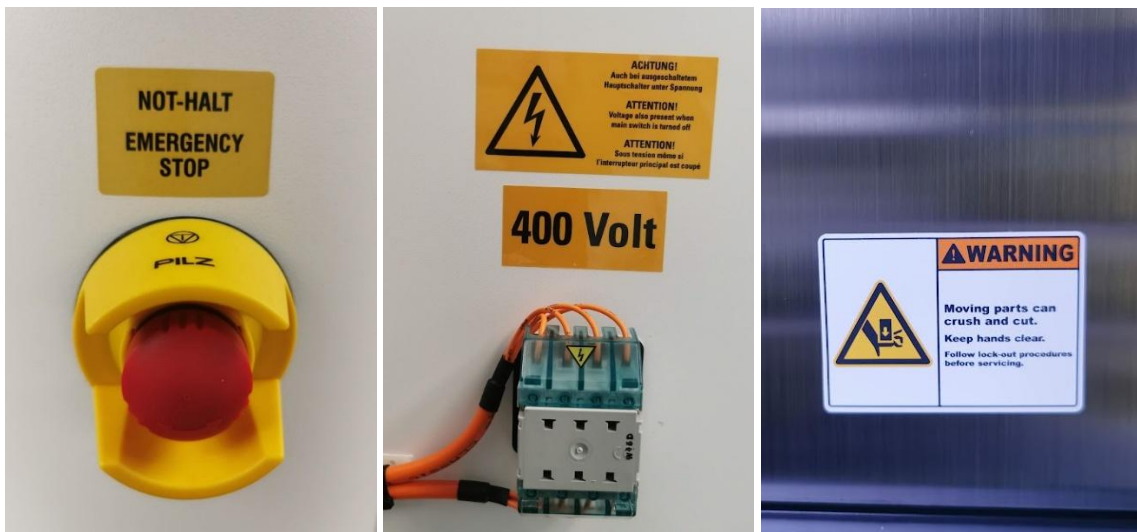
11.1 Visual Inspection

11.1.1 Check the Housing

Visually check the housing for damage

11.1.2 Check the Labeling

Visually check the labelling, e.g. safety label must be attached.



11.1.3 Check the Top of Machine

Visually check if there is any damage on top of the machine



11.2 Check FFU

11.2.1 Check for Loud Noises

Check the Filter Fan Units (FFU) for loud noises. Reduce the fan speed if necessary.

11.3 Ionizer

11.3.1 Check the Ionizer

Every two years, visually check the ionizer for damage.



11.3.2 Clean the Ionizer Tips

Clean the ionizer tips regularly



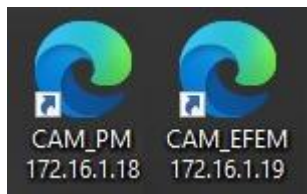
11.4 Check IP cameras



11.4.1 Check the IP cameras

Every two years, check both the EFEM and PM cams for performance.

1. Launch with the camera's shortcut on the desktop and login to the cameras with login ID "root" and password "1234".

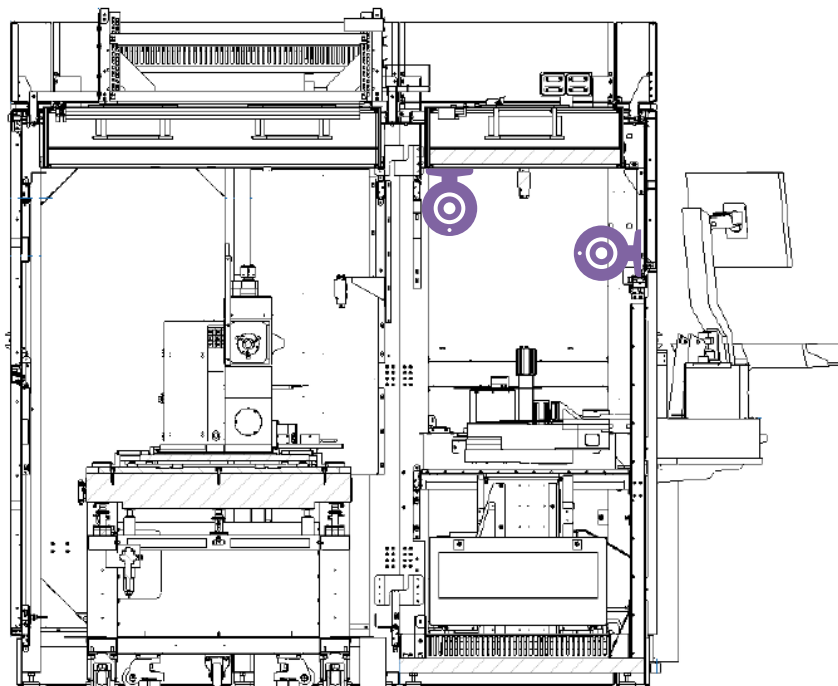


2. Check if the cameras are working and recording clips are available.

11.4.2 Exchange the SD card in the IP Cameras

Every two years, exchange the SD card in the IP camera.

1. The IP cams are located at EFEM area.



2. Remove the cover of IP camera and unscrew the screws at both sides of the camera using torx key 9 (Star shape).



3. Eject the SD card and insert the new SD card.



4. Screw back the two screws and place back the cover.
5. Login the camera and fine adjust the camera angle so that covering the views of interest (robot movement area, prealigner, loadports and stage of PM)

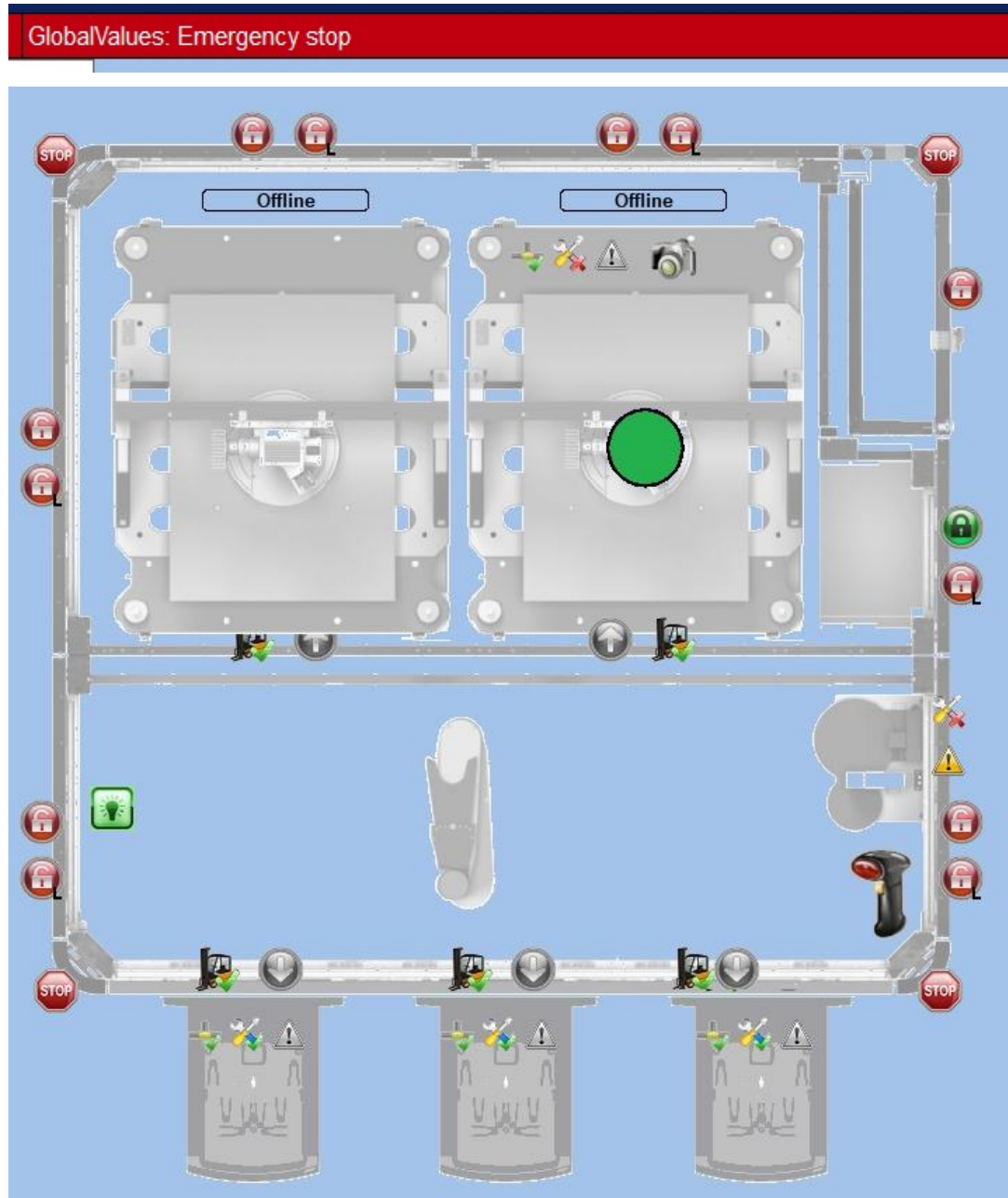
12 Safety Check

12.1 Safety Check

12.1.1 Perform Safety Check

Perform a complete safety check. Follow the Davinci - Safety Test instruction.

Example: 3 LPs, 2 PMs



Example: 2 LPs, 1 PM

